AN EXPLICIT LOWER BOUND IN TURÁN'S PURE POWER SUM PROBLEM

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ABSTRACT. For $n \geq 1$ let $R_n = \min \max_{1\leq k\leq n} |z_1^k + \cdots + z_n^k|$, the minimum being over all complex numbers $z_1 = 1, z_2, \ldots, z_n$. This is Turán's pure power sum problem (Erdős problem #519). In 2000 Biró proved that $R_n > q$ for every n, where $q > 1/2$ is an effectively computable absolute constant, but he did not compute a value of q. Following his proof step by step and making every constant in it explicit, we show that

$$R_n > \tfrac{1}{2} + 2^{-32\,176\,926\,823\,186\,696} \qquad \text{for every } n;$$

in particular $R_n > 1/2 + 10^{-10^{16}}$. The method is entirely Biró's. Inside his argument we add a few sharpenings, all of which are listed. With his own versions of the angle bound (Lemma 7) and of the final step, the same computation gives $R_n > 1/2 + 2^{-318\,600\,000\,000\,000\,000\,400}$. The final numerical inequalities are verified in ball arithmetic. A much better explicit value, $R_n > 1/2 + 2^{-4096}$, had already been proposed by R. Pinho, with a contraction argument in place of part of Biró's Lemma 8; our value is the one that Biró's argument gives as it is written. The work was done with substantial help from an AI model; it has not been refereed.

1. INTRODUCTION

For complex numbers $z_1, \ldots, z_n$ write $S_k = z_1^k + \cdots + z_n^k$ and put

$$R_n = \min_{z_1=1,\, z_2,\ldots,z_n\in\mathbb{C}} \max_{1\leq k\leq n} |S_k|.$$

Biró defines R_n with the normalization $\max_j |z_j| = 1$, and notes that the minimum exists by compactness and that this normalization can be replaced by $z_1 = 1$ [Bir00b, §1]. Turán conjectured that R_n is bounded below by a positive constant independent of n, and asked for the best such constant [Tur84]. This is Erdős problem #519 [Blo], and $\limsup R_n$ is the constant C_{42} of the Optimization Constants repository [Tao]. Atkinson proved the conjecture with the constant $1/6$ [Atk61] and later improved it to $\pi/8$ [Atk69]. Biró proved $R_n > 1/2$ [Bir94], and then the following.

Theorem (Biró [Bir00b]). *There is an effectively computable absolute constant $q > 1/2$ such that $\max_{1\leq j\leq n} |S_j| > q$ for every n and all complex numbers $z_1 = 1, z_2, \ldots, z_n$.*

Biró remarks that a concrete value of q could be obtained by following the steps of his proof, but he does not compute one [Bir00b, §1]. The entry for C_{42} in [Tao] records that this value is not known, and the forum page of Erdős problem #519 asks for it [Blo]. In the other direction, Biró proved $\limsup R_n < 1$ [Bir00]; see [Tao] for the current upper bounds. In this note we compute the explicit value that Biró's argument gives when it is followed as it is written.

Prior work. After the first version of this note was made public, we learned that R. Pinho had already proposed, on 2 November 2026, the much better explicit value $R_n > 1/2 + 2^{-4096}$ [Pin310] (unrefereed, prepared with AI). We had missed it. His argument and ours are compared in §4.4; our Theorems 1.1 and 1.2 are far weaker.

Theorem 1.1. *For every $n \geq 1$ and all complex numbers $z_1 = 1, z_2, \ldots, z_n$,*

$$\max_{1\leq k\leq n} |z_1^k + \cdots + z_n^k| > \frac{1}{2} + 2^{-N_0}, \qquad N_0 = 32\,176\,926\,823\,186\,696.$$

Date: November 2026 (version 1.0.2). Unrefereed.

2020 *Mathematics Subject Classification.* Primary 11N30; Secondary 30A10, 65G30.

Key words and phrases. Turán's pure power sum problem, power sums, Erdős problem #519, explicit constants, interval arithmetic.

Consequently $R_n > \frac{1}{2} + 2^{-N_0} > \frac{1}{2} + 10^{-10^{16}}$ *for every* n.

Theorem 1.2. *The conclusion of Theorem 1.1 also holds with* N_0 *replaced by*

$$N_1 = 318\,600\,000\,000\,000\,000\,400,$$

so that $R_n > \frac{1}{2} + 10^{-10^{20}}$*, by a proof that uses Biró's own Lemma 7 (*$\beta_t \le \pi/8$*) and his own final step (45)–(47).*

Credit. The method and all of its ideas are A. Biró's [Bir00b]. We follow his proof step by step, give an explicit admissible value to every implicit constant in it (the K_i, m_i, f_i, the O-terms and the conditions "large enough" or "small enough"), and verify the resulting finite list of inequalities rigorously at one choice of parameters. Inside his argument we add ten small changes S1–S10 that improve some constants; none of them changes the method. Section 4 lists every step that is not in [Bir00b], with proofs, and Table 1 shows which changes each of Theorems 1.1 and 1.2 uses.

Why the bound is so close to $1/2$**.** The following estimates are floating-point approximations at the parameters of Theorem 1.1; they are only a guide and are not used in any proof. Write $\delta = q - 1/2$. Three features of the argument determine the size of N_0.

(i) Lemma 8 of [Bir00b] shows that $h_k = |D_{k-1}|/C_{k-1}$ is close to 1 (notation in Section 2), but only at a logarithmic rate: $1 - h_k \le g_9 \approx 3K_{11}/\log(1/\delta)$ with $K_{11} \approx 0.108$. This is a feature of Biró's proof of Lemma 8, not of the method: Pinho's contraction argument gives a rate that is a power of δ (§4.4).

(ii) In (38) and (41) of [Bir00b] this deviation enters through square roots, so the final error term is $f_{15} \approx 1.93\sqrt{g_9}$.

(iii) Corollary 2 of [Bir00b] transfers smallness from the c_j to the b_j at a high price: its bound contains the term $(\varepsilon + 1/k)^{1/\sqrt{2}-q}$ with exponent $1/\sqrt{2} - q \approx 0.207$, which forces $\varepsilon \approx 10^{-4}$, and a term proportional to $f_{15}/(\gamma\varepsilon)$.

The final condition $f_{19} \le 1/4$ then needs $f_{15} \approx 7 \cdot 10^{-9}$, hence $g_9 \approx 1.5 \cdot 10^{-17}$, hence $\log(1/\delta) \approx 2 \cdot 10^{16}$. So, in Biró's argument as written, $q - 1/2$ is doubly exponentially small.

Status. This note has not been refereed, and no professional mathematician has checked it yet. The work was done with substantial help from an AI model. Separate instances of the model re-derived the constants independently and found no error; Section 6 describes what was checked and what was not.

Organization. Section 2 recalls Biró's notation and the structure of his proof. Section 3 states the explicit form of his theorem (Theorem 3.1) with all formulas, and deduces Theorems 1.1 and 1.2. Section 4 lists everything that is not in [Bir00b]. Section 5 describes the rigorous numerical verification, and Section 6 the checks done so far. Appendix A contains the complete proof of Theorem 3.1, line by line, in the order of [Bir00b].

2. Biró's argument in brief

Fix q with $1/2 < q < 1/\sqrt{2}$, and suppose, to get a contradiction, that $z_1 = 1, z_2, \dots, z_n$ satisfy

$$|S_j| \le q \qquad (1 \le j \le n). \tag{1}$$

Biró also fixes an auxiliary bound $q < q_0 < 1/\sqrt{2}$ (his (2)), which keeps his constants uniform as $q \to 1/2$. Since we work at one fixed q, we take $q_0 = q$.

Notation. We use Biró's notation, with short names for his sums. The numbers b_j are defined by

$$(z - z_2)\cdots(z - z_n) = z^{n-1} + b_1 z^{n-2} + \cdots + b_{n-1}, \qquad b_0 = 1,$$

and for $t \ge 0$

$$c_t = b_0 + b_1 + \cdots + b_t, \quad B_t = 1 + |b_1| + \cdots + |b_t|, \quad C_t = 1 + |c_1| + \cdots + |c_t|, \quad D_t = 1 + c_1 + \cdots + c_t,$$

so $c_0 = B_0 = C_0 = D_0 = 1$, and B_t is nondecreasing. The Newton–Girard formulas and $z_1 = 1$ give [Bir94]

$$S_t + b_1 S_{t-1} + \cdots + b_{t-1} S_1 = c_{t-1} - t b_t \qquad (1 \le t \le n-1), \tag{3}$$

$$S_n + b_1 S_{n-1} + \cdots + b_{n-1} S_1 = c_{n-1}, \tag{4}$$

and summing (3) gives Biró's new formulas

$$S_t + c_1 S_{t-1} + \cdots + c_{t-1} S_1 = 2D_{t-1} - t c_t \qquad (1 \le t \le n-1). \tag{14}$$

For complex w_1, w_2 let $a(w_1, w_2) = |\arg(w_2/w_1)| \in [0, \pi]$ if $w_1 w_2 \neq 0$, and $a(w_1, w_2) = \pi$ otherwise (with $\arg \in (-\pi, \pi]$). Biró's angles are

$$\alpha_t = a(c_{t-1}, b_t), \qquad \beta_t = a(D_{t-1}, c_t).$$

If $\alpha_t \le \pi/4$ for all $1 \le t \le n-1$, then (4) and (1) give $q \ge 1/\sqrt{2}$; so there is a least k with $\alpha_k > \pi/4$, and $1 \le k \le n-1$ (Appendix A.2):

$$\alpha_t \le \pi/4 \quad (1 \le t \le k-1), \qquad \alpha_k > \pi/4. \tag{5, 6}$$

The steps.

(a) *Lemmas 2–5 and Corollaries 1–2.* For $t \le k$ all earlier angles are at most $\pi/4$, so $|c_{t-1}| \ge B_{t-1}/\sqrt{2}$, while (3) and (1) give $|c_{t-1} - t b_t| \le q B_{t-1}$. As q is close to $1/2$, this forces near-equality: $|c_{t-1}| \approx B_{t-1}/\sqrt{2}$ (Lemma 4) and $t|b_t| \approx B_{t-1}/2$ outside a small set H_α of indices with $\alpha_t \le \alpha$ (Lemma 5). Moreover $B_{k-1} > M(q) = (\sqrt{2}-1)/(2(q-1/2))$, so k is large (Corollary 1). Corollary 2 turns smallness of $\sum_{t \in H} |c_t|$ into smallness of $\sum_{t \in H} |b_t|$.

(b) *Lemma 6.* $t|c_t|/C_{t-1} = 3/2 + \rho_t$ with $|\rho_t| \le f_5 B_{k-1}/B_t$.

(c) *Lemma 7.* By (14) and (1), $|2D_{t-1} - t c_t| \le q C_{t-1}$. This forces $\beta_t \le \pi/8$ for $t \le k$.

(d) *Lemma 8.* The ratio $h_t = |D_{t-1}|/C_{t-1}$ satisfies $h_{t+1} \ge h_t + K_5(\cos\beta_t - h_t)/t$ when $\cos\beta_t \ge h_t$ (18), and, up to a small error, $\cos\beta_t \ge F(h_t)$ (19), where $F(h) = (2h^2+1)/(3h)$. Since $F(h) > h$ for $0 < h < 1$, the ratio h_t is pushed towards the fixed point 1, and one gets $h_k \approx 1$ and $\cos\beta_k \approx 1$.

(e) *Section 4 of* [Bir00b]. By (14) at $t = k$ and (d), $\sum_j |t_j| S_{k-j} \approx 1/2$, where $t_j = c_j/D_{k-1}$ and $\sum_j |t_j| \approx 1$. Since $\operatorname{Re} S \le q \approx 1/2$, the indices j with $\operatorname{Re} S_{k-j} \le c := 1/2 - \gamma$ carry little weight, and for the other indices $S_{k-j} \approx 1/2$. Corollary 2 moves this to the weights b_j, and (3) at $t = k$ gives $k b_k \approx c_{k-1}/2$ (44). So b_k points almost in the direction of c_{k-1}, against $\alpha_k > \pi/4$.

Parameters. Besides q, the argument has the parameter $\alpha \in (0, \pi/4)$ of Lemma 5 (Biró's own), and in our explicit form the following auxiliary ones: $\beta_0 \in \{\pi/8, \frac{1}{2}\arcsin q\}$ (the angle bound used from Lemma 7 on), $\theta \in (0,1)$ and $\mu \in (0, 2/3)$ (Lemma 8; Biró uses $\theta = 1/3$, and μ plays the role of his $1/K_6$), $\varepsilon \in (0,1)$ (Corollary 2) and $\gamma > 0$ (the choice $c = 1/2 - \gamma$ in Section 4 of [Bir00b]).

3. The explicit theorem

Theorem 3.1 (explicit form of Biró's theorem)**.** *Let $q, \alpha, \beta_0, \theta, \mu, \varepsilon, \gamma$ be real numbers with*

(P1) $1/2 < q < 1/\sqrt{2}$;
(P2) $0 < \alpha < \pi/4$;
(P3) $\beta_0 = \pi/8$ *or* $\beta_0 = \frac{1}{2}\arcsin q$;
(P4) $0 < \theta < 1$, $0 < \mu < 2/3$, $0 < \varepsilon < 1$, $\gamma > 0$.

Define the numbers in steps 1–14 below, and assume

(L8a) $m_5 > 1$;
(L8b) $(2+q) f_8^\theta < 1$;
(FIN) $f_{19} \le 1/4$.

Then $\max_{1 \le j \le n} |S_j| > q$ *for every* $n \ge 1$ *and all complex numbers* $z_1 = 1, z_2, \ldots, z_n$. *The same conclusion holds if* (FIN) *is replaced by*

(FIN-B) $f_{19} \le (2 - \sqrt{2})/4$,

which implies (FIN)*; with* (FIN-B) *the proof uses Biró's final step* (45)–(47) *instead of Proposition 4.3.*

With $\delta := q - 1/2$ and log the natural logarithm, the numbers are:

1. $K_1 = 1/\sqrt{2} - q, \quad M = \dfrac{\sqrt{2}-1}{2\delta}, \quad \kappa = M^{-1/(1+q)}.$
2. $f_1 = \dfrac{\sqrt{2}\,\delta}{\cos\alpha - 1/\sqrt{2}}.$
3. $f_{2b} = \dfrac{q}{\sin\alpha} - \dfrac{1}{\sqrt{2}}, \quad f_{2a} = \sqrt{q^2 - \tfrac{1}{2}\sin^2\alpha} + \min\Big(1, \dfrac{q}{\sin\alpha}\Big)\Big(\cos\alpha - \dfrac{1}{\sqrt{2}}\Big) + \dfrac{f_{2b}}{\sqrt{2}}.$
4. $f_3 = f_{2a} + \dfrac{f_1}{2K_1}.$
5. $f_5' = (2+q)(f_3 + 3\delta), \quad f_5'' = (2+q)\big(f_3 + 3\delta + (\sqrt{2}(1+q) + \tfrac{1}{2})\kappa\big).$
6. $f_7 = q^2 - \tfrac{1}{4}, \quad f_8 = \dfrac{f_7 + f_5'}{6\cos\beta_0}, \quad f_8^{(k)} = \dfrac{f_7 + f_5''}{6\cos\beta_0}.$
7. $K_{11} = \dfrac{\sqrt{2}(1+q)(1-\cos\beta_0)}{2/3 - \mu}, \quad m_5 = \dfrac{M}{2+q}, \quad K_{12} = \dfrac{1}{2+q}.$
8. $f_9 = \max\Big\{\dfrac{f_8^{1-\theta}}{\mu}, \dfrac{K_{11}}{\log m_5}, \dfrac{K_{11}}{\log(K_{12} f_8^{-\theta})}\Big\}, \quad g_9 = \min(f_9, 1 - \cos\beta_0).$
9. $f_{10} = g_9/3 + f_8^{(k)}.$
10. $f_{11} = \tfrac{3}{2}g_9 + \tfrac{3}{2}\sqrt{2f_{10}} + f_5''.$
11. $f_{12} = \dfrac{g_9}{1-g_9}, \quad f_{13} = \dfrac{\sqrt{2g_9}}{1-g_9}.$
12. $f_{14} = \dfrac{f_{11}}{1-g_9} + q f_{13}, \quad f_{15} = \delta + q f_{12} + f_{14}.$
13. $y = f_{15}/\gamma, \quad f_{17} = \min\Big\{1,\ (\varepsilon+\kappa)^{K_1} + \sqrt{2}(1+q)\,\dfrac{y}{\varepsilon}\Big\}.$
14. $f_{18} = \sqrt{f_7 + \gamma}, \quad f_{19} = f_{18} + (q + \tfrac{1}{2})f_{17}.$

Under (P1), (P2), the quantities K_1, $\sin\alpha$, $\cos\alpha - 1/\sqrt{2}$, $q^2 - \frac{1}{2}\sin^2\alpha$ and f_{2b} are positive, and under (L8a), (L8b) both logarithms in f_9 are positive; so all numbers are well defined and $0 < g_9 < 1$. The numbered symbols f_i, K_i, m_i correspond to the constants and functions of [Bir00b] with the same index (Table 4 in Appendix A).

Proof. Appendix A, which follows [Bir00b] step by step; the theorem is assembled in Appendix A.26. □

Remark 3.2. Theorem 3.1 gives $R_n \geq q$, and in fact $R_n > q$ because the minimum defining R_n is attained [Bir00b, §1]. (If $|S_k| \leq 1$ for $k \leq n$, Newton's identities give $|e_k| \leq 1$ for the elementary symmetric functions of $z_1, \ldots, z_n$, so all $|z_j| \leq 2$; we minimize a continuous function over a compact set.)

Proof of Theorems 1.1 and 1.2. At the two parameter points of Table 2, with $q = \frac{1}{2} + 2^{-N_0}$ and $q = \frac{1}{2} + 2^{-N_1}$, the conditions (P1)–(P4), (L8a), (L8b) and (FIN) hold, and at the second point also (FIN-B). This is verified rigorously in Section 5. Theorem 3.1 then gives the first statement of both theorems (with $\beta_0 = \pi/8$ and (FIN-B) for Theorem 1.2), and Remark 3.2 gives $R_n > q$. Finally $N_0 \log_{10} 2 = 9\,686\,220\,142\,064\,131.31\ldots < 10^{16}$ and $N_1 \log_{10} 2 = 95\,908\,156\,618\,544\,408\,915.5\ldots < 10^{20}$ (checked in ball arithmetic), so $2^{-N_0} > 10^{-10^{16}}$ and $2^{-N_1} > 10^{-10^{20}}$. □

4. What is not in Biró's paper

Appendix A follows [Bir00b] step by step, except for the points below: one misprint (§4.1), ten changes inside the argument (§4.2), and some bookkeeping that is not a new idea (§4.3).

None of them changes the method. We write $(\star)$ for the inequality

$$(\star) \qquad |c_{t-1}| \ge \Big(1 - \frac{1}{\sqrt{2}}\Big) + \frac{B_{t-1}}{\sqrt{2}} \qquad (1 \le t \le k),$$

which follows from Lemma 2(ii) of [Bir00b] and (5) (Appendix A.3).

4.1. **A misprint.** In formula (18) of [Bir00b] the left-hand side h_{t-1} should read h_{t+1}: the display just before (18) bounds h_{t+1}, and (18) is used to go from t to $t+1$. Nothing else depends on this. Both independent checks of Section 6 found it as well.

4.2. **The changes S1–S10.**

S1. Lemma 7 with the angle $\beta^* = \frac{1}{2}\arcsin q$. Biró's Lemma 7 states $\beta_t \le \pi/8$ for $1 \le t \le k$. We use the following form; part (a) with $\beta = \pi/8$ is his statement, and his proof gives (a) without change.

Lemma 4.1 (Lemma 7 of [Bir00b], general form)**.** *Assume* (1) *with* $1/2 < q < 1/\sqrt{2}$.

(a) *If* $\beta \in (0, \pi/2)$ *and* $\sin 2\beta > q$, *then* $\beta_t \le \beta$ *for* $1 \le t \le k$.

(b) $\beta_t \le \beta^* := \frac{1}{2}\arcsin q$ *for* $1 \le t \le k$.

Proof. (a) Suppose not, and let t be the least index in $[1, k]$ with $\beta_t > \beta$. Then $\beta_j \le \beta$ for $1 \le j \le t-1$.

(15) $|D_{t-1}| \ge C_{t-1}\cos\beta$. For $1 \le j \le t-1$, Lemma 1′(iv′) (Appendix A.1) with $w_1 = D_{j-1}$, $w_2 = c_j$ gives $|D_j| \ge |D_{j-1}| + |c_j|\cos\beta_j \ge |D_{j-1}| + |c_j|\cos\beta$. With $|D_0| = 1$ this gives $|D_{t-1}| \ge 1 + (C_{t-1} - 1)\cos\beta \ge C_{t-1}\cos\beta$. In particular $D_{t-1} \ne 0$.

(16) $|2D_{t-1} - tc_t| \ge 2|D_{t-1}|\sin\beta$. If $\beta_t \le \pi/2$, Lemma 1′(iii′) with $w_1 = 2D_{t-1}$, $w_2 = tc_t$ (so that $a(w_1, w_2) = \beta_t$) gives $|2D_{t-1} - tc_t| \ge 2|D_{t-1}|\sin\beta_t$, and $\sin\beta_t > \sin\beta$ because $\beta < \beta_t \le \pi/2$. If $\beta_t > \pi/2$ and $c_t = 0$, the left side equals $2|D_{t-1}|$. If $\beta_t > \pi/2$ and $c_t \ne 0$, then $a(2D_{t-1}, -tc_t) = \pi - \beta_t < \pi/2$, and Lemma 1′(iv′) gives $|2D_{t-1} - tc_t| \ge 2|D_{t-1}| + t|c_t|\cos(\pi - \beta_t) \ge 2|D_{t-1}|$.

Combining (15) and (16), $|2D_{t-1} - tc_t| \ge 2\sin\beta\cos\beta\, C_{t-1} = \sin(2\beta)\, C_{t-1}$. By (14) and (1) (valid since $t \le k \le n-1$), the left side is at most qC_{t-1}, and $C_{t-1} \ge 1$. So $\sin 2\beta \le q$, a contradiction.

(b) Let $\beta \in (\beta^*, \pi/4]$. Then $2\beta \in (\arcsin q, \pi/2]$, and since sin is strictly increasing on $[0, \pi/2]$, $\sin 2\beta > \sin(\arcsin q) = q$. By (a), $\beta_t \le \beta$. So $\beta_t \le \beta$ for every $\beta \in (\beta^*, \pi/4]$, hence $\beta_t \le \inf(\beta^*, \pi/4] = \beta^*$. □

One cannot simply put $\beta = \beta^*$ into (a): then $\sin 2\beta = q$ and the argument ends with $q \le q$; this is why (b) goes through the limit. Since $1/2 < q < 1/\sqrt{2}$ we have $\pi/12 < \beta^* < \pi/8$, so every later use of Lemma 7 in [Bir00b] (through $\cos\beta_t \ge \cos\beta_0$ and $\cos^2\beta_0 > 1/2$) remains valid with $\beta_0 = \beta^*$. The gain is in K_{11}, which is proportional to $1 - \cos\beta_0$: $1 - \cos\beta^* \approx 0.034$ against $1 - \cos(\pi/8) \approx 0.076$ (floats).

S2. Formula (19) with an error linear in X_t and coefficient 1. Let $h_t = |D_{t-1}|/C_{t-1}$, $X_t = B_{k-1}/B_t$, $F(h) = (2h^2+1)/(3h)$ and $\rho_t = t|c_t|/C_{t-1} - 3/2$. By (17) (Appendix A.11), $\cos\beta_0 \le h_t \le 1$ for $1 \le t \le k$, and by Lemma 6 (Appendix A.9) $|\rho_t| \le f_5' X_t$ for $t \le k-1$ and $|\rho_k| \le f_5'' X_k$. Biró's (19) reads $\cos\beta_t \ge F(h_t) + O(f_8 X_t^2)$.

Proposition 4.2 ((19) of [Bir00b], sharper form)**.** *For* $1 \le t \le k$,

$$(19') \qquad \cos\beta_t \;\ge\; F(h_t) - \frac{f_7 + |\rho_t|}{6h_t} \;\ge\; F(h_t) - \frac{f_7 + |\rho_t|}{6\cos\beta_0}, \qquad f_7 = q^2 - \tfrac{1}{4}.$$

Consequently $\cos\beta_t \ge F(h_t) - f_8 X_t$ *for* $1 \le t \le k-1$, *and* $\cos\beta_k \ge F(h_k) - f_8^{(k)}$.

Proof. Fix t and write $h = h_t$, $x = \cos\beta_t$, $\rho = \rho_t$. By Lemma 4.1, $\beta_t \le \beta_0 < \pi$, so $D_{t-1} \ne 0$, $c_t \ne 0$, and $x \ge \cos\beta_0 > 1/\sqrt{2}$. Put $w_1 = 2D_{t-1}/C_{t-1}$ and $w_2 = tc_t/C_{t-1}$. Then $|w_1| = 2h$, $|w_2| = 3/2 + \rho$, and $\varphi = \arg(w_1/w_2) = \arg(D_{t-1}/c_t)$ satisfies $|\varphi| = \beta_t$, so $\cos\varphi = x$. By (14) and (1), $|w_1 - w_2| \le q$. Lemma 1(ii) of [Bir00b] gives

$$4h^2 + (3/2+\rho)^2 - 4h(3/2+\rho)x = |w_1 - w_2|^2 \le q^2,$$

which is equivalent to

$$6hx \geq 4h^2 + 2 - f_7 + T, \qquad T := 3\rho + \rho^2 - 4h\rho x.$$

We claim $T \geq -|\rho|$. If $\rho \geq 0$, then $T + \rho = \rho^2 + 4\rho(1 - hx) \geq 0$, since $0 < h \leq 1$ and $x \leq 1$. If $\rho < 0$, put $r = -\rho$; then $T + r = r^2 + r(4hx - 2) \geq 0$, since $h \geq \cos\beta_0$ and $x \geq \cos\beta_0$ give $4hx \geq 4\cos^2\beta_0 > 2$. Hence $6hx \geq 4h^2 + 2 - f_7 - |\rho|$, and dividing by $6h > 0$ gives the first inequality of (19′). The second follows from $h \geq \cos\beta_0$ and $f_7 + |\rho| \geq 0$. For $t \leq k - 1$ we have $X_t \geq 1$, so $f_7 + |\rho_t| \leq (f_7 + f_5')X_t$; for $t = k$ we have $X_k \leq 1$, so $|\rho_k| \leq f_5''$. □

Compared with [Bir00b], the error is linear in X_t (the term ρ^2 has the good sign and is dropped), and $|\rho_t|$ has coefficient 1 (the cross term $-4h\rho x$ is controlled by Lemma 7 and (17); a crude bound would give 7). Biró's f_6 becomes simply f_5. The linearity is what allows S6.

S3. Formula (20) with $2(1 - H)/3$. For $0 < H \leq 1$,

$$F(H) - H = \frac{1 - H^2}{3H} \;\geq\; \frac{2(1 - H)}{3}, \tag{20′}$$

because with $H = 1 - \varphi$ one has $F(H) - H - 2\varphi/3 = \varphi^2/(3(1 - \varphi)) \geq 0$. Biró's (20) uses $(1 - H)/3$. As a result, the gap in his (26) becomes $\cos\beta_t - h_t \geq K_7\varphi$ with $K_7 = 2/3 - \mu$ (Appendix A.15), and K_{11} is roughly halved.

S4. Formula (27) with $h_{t_0} \geq \cos\beta_0$. In (27) Biró bounds the total increase $h_k - h_{t_0}$ using $h_{t_0} \geq 0$. By (17), $h_{t_0} \geq \cos\beta_0$, so the increase is at most $1 - \cos\beta_0$ instead of 1. This gives $K_9 = \sqrt{2}(1 - \cos\beta_0)/K_7$ (Appendix A.16).

S5. Formula (18) with the factor $1/(\sqrt{2}(t + 1))$. For $1 \leq t \leq k - 1$,

$$\frac{|c_t|}{C_t} \;\geq\; \frac{1}{\sqrt{2}\,(t + 1)}, \tag{18b}$$

since $(\star)$ at index $t + 1 \leq k$ gives $|c_t| \geq B_t/\sqrt{2}$, and $C_t \leq (t + 1)B_t$. Hence (18) holds in the form $h_{t+1} \geq h_t + (\cos\beta_t - h_t)/(\sqrt{2}(t + 1))$ whenever $\cos\beta_t \geq h_t$. Biró's form with K_5/t holds with $K_5 = 1/(2\sqrt{2})$; ours is about twice as strong for large t (Appendix A.13).

S6. A free exponent θ *in the choice of* t_0. Biró chooses t_0 in (29)–(32) with the threshold $f_8^{-1/3}$. We use $f_8^{-\theta}$ with any $\theta \in (0, 1)$. With the quadratic error of Biró's (19) one needs $\theta < 1/2$; with (19′) every $\theta < 1$ works. Since $f_9 \approx K_{11}/(\theta \log(1/f_8))$ (float), θ close to 1 gains a factor of about 3 over $\theta = 1/3$ (Appendix A.17).

S7. A free γ *in the choice of* c. In Section 4 of [Bir00b], Biró takes $c = 1/2 - f_{16}$ with $f_{16} = \sqrt{f_{15}}$. We take $c = 1/2 - \gamma$ with a free $\gamma > 0$ and $y = f_{15}/\gamma$. This balances the two parts of the final bound: f_{17} wants γ large, $f_{18} = \sqrt{f_7 + \gamma}$ wants γ small (Appendix A.22).

S8. The final step with threshold $1/4$. Biró's final step (45)–(47) uses (6) only through $\sin\alpha_k > 1/\sqrt{2}$ (his Lemma 2(iii)), and gives a contradiction if $f_{19} \leq (2 - \sqrt{2})/4 \approx 0.146$. The following uses the angle condition (6) directly.

Proposition 4.3. *If* $f_{19} \leq 1/4$, *then* (44′) *of Appendix A.25, that is* $|c_{k-1}/2 - kb_k| \leq f_{19}B_{k-1}$, *together with* $(\star)$ *contradicts* (6).

Proof. We have $c_{k-1} \neq 0$ by $(\star)$ and $b_k \neq 0$ (Appendix A.3). Put $e = (kb_k - c_{k-1}/2)/c_{k-1}$, so that $kb_k/c_{k-1} = 1/2 + e$. By (44′), by $(\star)$ in the strict form $|c_{k-1}| > B_{k-1}/\sqrt{2}$, and since $f_{19}B_{k-1} > 0$,

$$|e| \leq \frac{f_{19}B_{k-1}}{|c_{k-1}|} < \sqrt{2}\, f_{19} \leq \frac{\sqrt{2}}{4} = \frac{1}{2\sqrt{2}}.$$

Let $w = 1/2 + e$. Then $\operatorname{Re} w \geq 1/2 - |e| > 0$. Since $(a + b)^2 \leq 2(a^2 + b^2)$ for real a, b, we have $|\operatorname{Im} e| - \operatorname{Re} e \leq \sqrt{2}|e|$, hence

$$|\operatorname{Im} w| - \operatorname{Re} w = |\operatorname{Im} e| - \operatorname{Re} e - \tfrac{1}{2} \leq \sqrt{2}|e| - \tfrac{1}{2} < 0.$$

So $w \neq 0$ and $|\arg w| < \pi/4$. As $k > 0$, $\alpha_k = |\arg(b_k/c_{k-1})| = |\arg w| < \pi/4$, contradicting (6). □

The threshold $1/4$ is the best that (44′), ($\star$) and (6) alone can give: the largest disc around $1/2$ inside the sector $|\arg w| < \pi/4$ has radius $1/(2\sqrt{2})$. In floats, this change reduces the exponent from about $1.39 \cdot 10^{20}$ (with S1 and Biró's final step) to $N_0 \approx 3.22 \cdot 10^{16}$.

S9. Formula (12) from Lemma 4. For $1 \le t \le k$,

$$0 \le C_{t-1} - \frac{1}{\sqrt{2}} \sum_{j=1}^{t} B_{j-1} \le \sqrt{2}\,(q - \tfrac{1}{2})\, t\, B_{k-1}, \tag{12′}$$

by adding Lemma 4 of [Bir00b] in the explicit form $0 \le |c_{j-1}| - B_{j-1}/\sqrt{2} \le \sqrt{2}(q-\frac{1}{2})B_{k-1}$ over $j = 1, \dots, t$. Biró obtains (12) from Lemma 5, treating $j \in H_\alpha$ and $j \notin H_\alpha$ separately. Our route is shorter and also covers $t = k$ (Appendix A.9).

S10. The function m_2 via Bernoulli's inequality. Biró's $m_2(k) = \prod_{l=1}^{k-1}(1 + K_2/l)$ bounds B_{k-1} from above. We use $B_{k-1} \le k^{1+q}$: by Lemma 3(i), $B_t/B_{t-1} \le 1 + (1+q)/t \le (1 + 1/t)^{1+q}$ (Bernoulli's inequality, exponent ≥ 1), and the product over $t = 1, \dots, k-1$ telescopes. With Corollary 1 this gives $k > M^{1/(1+q)}$, i.e. $1/k < \kappa$ (Appendix A.4, A.5).

4.3. **Bookkeeping that is not a new idea.**

(B1) $q_0 = q$ (Section 2).

(B2) Degenerate angles: Lemma 1′ states Biró's Lemma 1(iii), (iv) with the convention that $a(w_1, w_2) = \pi$ when $w_1 w_2 = 0$ (Appendix A.1).

(B3) Lemma 3: we keep the constants of parts (i) and (ii) separate: $K_1^{(\mathrm{i})} = 1/\sqrt{2} - q$ and $K_2^{(\mathrm{i})} = 1 + q$ for (i), and $K_1^{(\mathrm{ii})}$, $K_2^{(\mathrm{ii})}$ for (ii). Biró uses one pair for both ("perhaps changing the constants"). In particular K_{10}, K_{12}, m_2 and m_5 use only the constant $K_2^{(\mathrm{i})} = 1 + q$ of part (i).

(B4) Corollary 2 is stated with a free $\varepsilon \in (0, 1)$, and $1/k < \kappa$ is taken from Corollary 1; Biró instead argues that k may be assumed arbitrarily large. Our form needs no smallness assumption on y.

(B5) Steps that [Bir00b] states in one sentence are written out: Lemma 6 at $t = k$ (the extra term $(\sqrt{2}(1+q) + \frac{1}{2})\kappa$ in f_5''), (19) at $t = k$, the trivial branches of Lemma 8 (the constants K_{13}–K_{15} are replaced by the three-term maximum f_9 and by $g_9 = \min(f_9, 1 - \cos\beta_0)$), and the steps (38)–(47) with every constant.

4.4. **Comparison with Pinho's argument.**

R. Pinho [Pin310] replaces the second half of Lemma 8 (the threshold $H = 1 - \varphi$, Lemma 8a and the harmonic sum in (27)) by a contraction. In our notation, put $g_t = 1 - h_t$ and $w_t = |c_t|/C_t$, and let E bound the error term in (19) for $t_0 \le t \le k$. The projection inequality (18a) and $F(h) - h \ge (1-h)/3$ give

$$g_{t+1} \le \Big(1 - \frac{w_t}{3}\Big) g_t + 2 w_t E,$$

and with $w_t \ge 1/t$ this yields $1 - h_k \le 6E + \prod_{t=t_0}^{k-1}(1 - \frac{1}{3t}) \le 6E + (t_0/k)^{1/3}$. So the deficit $1 - h_k$ decays like a power of k/t_0, and not like $1/\log(k/t_0)$ as in Lemma 8b. Since k/t_0 can be taken to be a power of $1/\delta$, the logarithmic loss that dominates our Theorems 1.1 and 1.2 disappears, and Pinho obtains $R_n > 1/2 + 2^{-4096}$ for every n. In his note (`LOWER_PROOF.md` in the folder of [Pin310]), the contraction is formulas (22)–(24) of his Section 5, and t_0 is chosen in his Section 4. His Sections 6–7 also appear to replace Biró's Corollary 2 and the final step by a different argument. His note was written and checked with AI models and has not been refereed. The author has not checked it; the comparison above was made with AI help and has not been verified independently.

Our Theorems 1.1 and 1.2 are therefore far from the best explicit value. Their only interest is as an explicit version of Biró's argument as it is written, with each of his constants identified

(Table 4). Pinho's contraction could be combined with the sharper constants of this note; we have not done this.

4.5. **Which changes each theorem uses.** Table 1 lists, for each proof, which of S1–S10 it uses. The independent derivation is described in Section 6; it uses Biró's route for (12) and the bound $B_{k-1} \le k(k+1)/2$ instead of S10.

TABLE 1. Changes used by each proof (+ used, – not used).

	Theorem 1.1	Theorem 1.2, our proof*	Theorem 1.2, independent derivation
S1 $\beta^* = \frac{1}{2}\arcsin q$	+	–	–
S2 (19′)	+	+	+
S3 (20′)	+	+	+
S4 $h_{t_0} \ge \cos\beta_0$	+	+	+
S5 (18b)	+	+	+
S6 free θ	+	+	+
S7 free γ	+	+	+
S8 threshold 1/4	+	–	–
S9 (12′)	+	+	–
S10 $B_{k-1} \le k^{1+q}$	+	+	–

*Theorem 3.1 with $\beta_0 = \pi/8$ and (FIN-B).

5. How the numbers were certified

Parameters. Table 2 gives the two parameter points. All inputs are exact: $q = \frac{1}{2} + \delta$ is a rational number, and α enters only through $x = \pi/4 - \alpha$, which is given exactly. At the first point $x = 2^{-\lfloor 2N_0/3 \rfloor}$.

TABLE 2. The two parameter points.

	Theorem 1.1	Theorem 1.2
$\delta = q - \frac{1}{2}$	$2^{-32176926823186696}$	$2^{-318600000000000000400}$
$x = \frac{\pi}{4} - \alpha$	$2^{-21451284548791130}$	$2^{-212400000000000000200}$
β_0	$\frac{1}{2}\arcsin q$	$\pi/8$
θ	$1 - 2^{-40}$	$1 - 102/106200000000000000101$
μ	2^{-30}	$1/100000$
ε	$12/100000$	$1/110000$
γ	$408/100000$	$1/714$
final test	(FIN): $f_{19} \le 1/4$	(FIN-B): $f_{19} \le (2-\sqrt{2})/4$

Arithmetic. The checker `check_eta.py` computes steps 1–14 of Section 3 in Arb ball arithmetic [Joh17], through python-flint 0.9.0 [FLINT]. Every quantity is a ball (midpoint and radius) that is guaranteed to contain the exact value, and the elementary functions (log, exp, $\sqrt{\ }$, sin, cos, arcsin, min, max) are evaluated with rigorous enclosures. A condition counts as satisfied only if it holds for every point of the ball; an undecided comparison counts as a failure. Condition (P4) is decided in exact rational arithmetic. Exponents such as $2^{-3\cdot 10^{20}}$ are handled natively by Arb.

Since δ and x are extremely small, the formulas are evaluated in a form that never subtracts nearly equal numbers. With $x = \pi/4 - \alpha$:

$$\sin\alpha = \frac{\cos x - \sin x}{\sqrt{2}}, \qquad \cos\alpha - \frac{1}{\sqrt{2}} = \frac{\sin x - 2\sin^2(x/2)}{\sqrt{2}},$$

$$q^2 - \tfrac{1}{2}\sin^2\alpha = \delta + \delta^2 + \tfrac{1}{4}\sin 2x, \qquad f_{2b} = \frac{\delta + \sin^2(x/2) + \frac{1}{2}\sin x}{\sin\alpha},$$

$$f_7 = \delta(1+\delta), \qquad M = \frac{\sqrt{2}-1}{2\delta}, \qquad 1 - \cos\beta_0 = 2\sin^2(\beta_0/2).$$

These identities were verified symbolically; δ and x themselves are inputs and are never recovered by subtraction.

Results.

- At the point of Theorem 1.1 the checker prints PASS at working precisions 64, 128, 256, 512 and 2000 bits. At 256 bits, $f_{19} = 0.24999999999999999903$ (rounded midpoint; the radius of the ball is below 10^{-30}), so the margin $1/4 - f_{19} \approx 9.7 \cdot 10^{-19}$ is far larger than the radius. The same output was obtained on a second computer (Windows, python-flint 0.9.0).
- With $N_0 - 1$ in place of N_0 (and the same rule $x = 2^{-\lfloor 2N/3 \rfloor}$, which gives the same x) the checker prints FAIL: $f_{19} = 0.25000000000000002577\ldots > 1/4$. So, for these values of $\theta, \mu, \varepsilon, \gamma$ and this rule for x, the exponent N_0 is sharp. Other choices of these parameters could lower it only in about the sixth significant digit (float search).
- At the point of Theorem 1.2 the checker prints PASS, and (FIN-B) holds. Here $f_{19} = 0.1462296737845\ldots$ and $(2-\sqrt{2})/4 = 0.1464466094067\ldots$, so the margin is about $2.2 \cdot 10^{-4}$.
- An independent floating-point prototype (mpmath, 60 digits; not a proof) agrees with all 21 compared values at the first point to relative accuracy $2.3 \cdot 10^{-46}$.
- The second checker of Section 6, written independently from [Bir00b] and [Bir94], prints PASS at its own best point, which has the same δ and x as the point of Theorem 1.2 (run by the author with python-flint 0.9.0, and reproduced separately). There its final value is $f_{19} = 0.1462564428\ldots$, also below $(2-\sqrt{2})/4$.

TABLE 3. Main values at the two points (ball midpoints, rounded).

	Theorem 1.1	Theorem 1.2
f_8	$6.200 \cdot 10^{-3228740047354711}$	$5.030 \cdot 10^{-31969385539514802963}$
K_{11}	0.10842	0.24222
$\log(K_{12} f_8^{-\theta})$	$7.434 \cdot 10^{15}$	$7.361 \cdot 10^{19}$
$f_9 = g_9$ (third term)	$1.4584 \cdot 10^{-17}$	$3.2905 \cdot 10^{-21}$
f_{15}	$7.3775 \cdot 10^{-9}$	$1.1082 \cdot 10^{-10}$
$(\varepsilon+\kappa)^{K_1}$	0.15416	0.09034
$\sqrt{2}(1+q)y/\varepsilon$	0.03197	0.01846
f_{17}	0.18613	0.10881
f_{18}	0.06387	0.03742
f_{19}	0.24999999999999999903	0.14622967378
threshold	1/4	$(2-\sqrt{2})/4 = 0.14644660941$

How the parameters were chosen. The parameters were found by a floating-point search; this is a guide only, and the verdict above does not depend on it. Write $L = \log(1/\delta)$ and $\Lambda = \log(1/x)$. Since $f_3 \approx \sqrt{x/2} + \delta/(K_1 x)$, the best ratio is $\Lambda \approx \frac{2}{3}L$, which gives $\log(1/f_8) \approx L/3$. In f_9 the third term dominates, $f_9 \approx 3K_{11}/(\theta L)$; the first term is made negligible by $1-\theta = 2^{-40}$ and $\mu = 2^{-30}$ (it is about 10^{-2928}). Then $f_{15} \approx 1.932\sqrt{g_9}$, and ε, γ minimize $f_{19} \approx \sqrt{\gamma} + \varepsilon^{K_1} + T$ with $T = \sqrt{2}(1+q)f_{15}/(\gamma\varepsilon)$. At the optimum $\sqrt{\gamma} = 2T$ and $\varepsilon^{K_1} = T/K_1$, and the certified point agrees with this. The first term $\varepsilon^{K_1} \approx 0.154$ of f_{17} is the largest single piece of the final budget $1/4$.

6. Status and independent checks

- *Unrefereed.* This note has not been refereed, and no professional mathematician has checked it. Until an expert has read it, it should not be regarded as established.
- *How it was made.* The derivation of all constants, all proofs and all code were produced with an AI model (Claude Opus 5.5), under the direction of the author.
- *First independent check.* A separate instance of the model, given only [Bir00b], [Bir94] and a description of the task, without access to our derivations, our checker or our parameter values, made every constant of [Bir00b] explicit and wrote its own checker. After renaming of parameters, its list of conditions corresponds one-to-one to ours. Every constant is either the same as ours or a different bound with its own valid proof; all of these proofs were re-read line by line (again by the model), and no error was found on either side. Its checker passes at its own points and fails where it should. At its best point our checker, run with $\beta_0 = \pi/8$ and (FIN-B), also passes, and the main intermediate values of the two chains agree to within 0.3%. It found the misprint of §4.1 independently. Its derivation also uses S2–S7: it listed S2, S3 and S6 as its own sharpenings, and used S4, S5 and S7 without listing them. It did not use S1, S8, S9 or S10 (Table 1).
- *Second independent check.* A further fresh instance, given only [Bir00b] and the two claims, first proved Lemma 4.1(b) and Proposition 4.3 on its own and then read our proofs. It found no error. At its suggestion the limit step in the proof of Lemma 4.1(b) is written out.
- *Not derived independently.* Four of our steps were only re-read by the model, not re-derived independently: S9, S10, the splitting used for f_{11} in $(38')$, and the case $(29')$ in the choice of t_0 (the first check avoided this case by its choice of threshold). In each of these places the first check proved a bound of the same kind by another route.
- *Test on real systems.* Theorem 3.1 cannot be tested directly, but every intermediate inequality of Appendix A holds for every system with $\max_j|S_j| \le q$ and any $q \in (1/2, 1/\sqrt{2})$. Near-optimal systems with $\max_j|S_j|$ between 0.688 and 0.707 exist for $n \ge 40$ [CG96, Cert]. On eleven such systems ($40 \le n \le 1000$, with $10 \le k \le 243$) we evaluated both sides of 50 of these inequalities and identities with our explicit constants (mpmath, 200 digits), including those of S1, S2, S5, S9, S10 and S8 (in the contrapositive form $|c_{k-1}/2 - kb_k| > B_{k-1}/4$). None was violated. This test cannot see errors that appear only when q is very close to $1/2$, for example in the main branch of Lemma 8.
- *Computer algebra.* The purely algebraic identities used in Appendix A were verified exactly with sympy. The inequalities were checked only by reading.
- *Limits of these checks.* All derivations were made by instances of one AI model, so they may share a blind spot. The ball arithmetic verifies the inequalities of Theorem 3.1 at the stated parameters; it does not verify Theorem 3.1 itself. Corrections are very welcome.

Use of AI. This work was carried out with substantial help from an AI model, Claude Opus 5.5 (Anthropic). It was used to read [Bir00b] and [Bir94], to derive and write the explicit constants and their proofs, to write all of the code, and to draft this text. The author planned and directed the work and ran the checkers on his own computer. The independent checks described above were done by separate instances of the same model.

Data and code. The checker `check_eta.py` with its complete outputs at the points of Theorems 1.1 and 1.2, the independently written second checker with the notes of its author, the floating-point cross-checks and the computer-algebra scripts are in the repository [Rep].

Acknowledgements. I thank András Biró for asking which explicit constant the proof of [Bir00b] gives, and Gergely Harcos, who had asked himself the same question.

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Appendix A. The proof of Theorem 3.1, step by step

This appendix proves Theorem 3.1. It follows [Bir00b] in order and keeps Biró's numbering of lemmas and formulas; our variants carry a prime, as in (19′). For every implicit constant of [Bir00b] it gives an explicit admissible value with a proof; Table 4 is a summary. The purely algebraic identities below were verified exactly with sympy; the inequalities are proved in the text.

Throughout, $1/2 < q < 1/\sqrt{2}$, the system $z_1 = 1, z_2, \dots, z_n$ satisfies (1), and $0 < \alpha < \pi/4$, so that $\cos\alpha > 1/\sqrt{2} > \sin\alpha > 0$. We write

$$\delta := q - \tfrac{1}{2}, \qquad \lambda := \sqrt{2}\,\delta,$$

and use the notation of Section 2. Biró's auxiliary bound q_0 is taken equal to q.

A.1. **Lemma 1′: degenerate angles.** Recall $a(w_1, w_2) = |\arg(w_2/w_1)|$ if $w_1 w_2 \neq 0$ and $a(w_1, w_2) = \pi$ otherwise. For all complex w_1, w_2:

(iii′) $|w_1 - w_2| \geq |w_1| \sin a(w_1, w_2)$;
(iv′) $|w_1 + w_2| \geq |w_1| + |w_2| \cos a(w_1, w_2)$.

Proof. If $w_1 w_2 \neq 0$, let $\varphi = \arg(w_1/w_2)$. Then $|\varphi| = a(w_1, w_2)$ (if $\arg z \neq \pi$ then $\arg(1/z) = -\arg z$, and if $\arg z = \pi$ then $\arg(1/z) = \pi$). Lemma 1(i) of [Bir00b] gives $|w_1 - w_2|^2 \geq |w_1|^2 \sin^2\varphi$, so $|w_1 - w_2| \geq |w_1||\sin\varphi| = |w_1| \sin a$; Lemma 1(iv) of [Bir00b] gives (iv′), since $\cos\varphi = \cos a$. If $w_1 w_2 = 0$, then $a = \pi$, and the claims read $|w_1 - w_2| \geq 0$ and $|w_1 + w_2| \geq |w_1| - |w_2|$, which always hold. □

We also use that $|\arg(-w)| = \pi - |\arg w|$ for $w \neq 0$ (if $\arg w = \vartheta \in (-\pi, \pi]$, then $\arg(-w) = \vartheta - \pi$ for $\vartheta > 0$ and $\vartheta + \pi$ for $\vartheta \leq 0$). Multiplying a number by a positive real changes neither its argument nor whether it is 0; so $\alpha_t = a(c_{t-1}, b_t) = a(c_{t-1}, t b_t)$.

A.2. **Setup.** $n \geq 2$. If $n = 1$, then $S_1 = z_1 = 1 > q$, contradicting (1).

k exists and $1 \leq k \leq n-1$. Suppose $\alpha_t \leq \pi/4$ for every $1 \leq t \leq n-1$. For each such t, Lemma 1′(iv′) with $w_1 = c_{t-1}$, $w_2 = b_t$ gives $|c_t| \geq |c_{t-1}| + |b_t| \cos\alpha_t \geq |c_{t-1}| + |b_t|/\sqrt{2}$. Starting from $|c_0| = 1$ and adding up, $|c_{n-1}| \geq 1 + (B_{n-1} - 1)/\sqrt{2} \geq B_{n-1}/\sqrt{2}$. On the other hand, (4)

TABLE 4. Biró's constants and our explicit values.

[Bir00b]	explicit value	where
K_1, K_2, Lemma 3(i)	$K_1^{(\mathrm{i})} = 1/\sqrt{2} - q$, $K_2^{(\mathrm{i})} = 1 + q$	A.4
K_1, K_2, Lemma 3(ii)	$K_1^{(\mathrm{ii})} = 1/(\sqrt{2}(1+q))$, $K_2^{(\mathrm{ii})} = 1 + 1/(1/\sqrt{2} - q)$	A.4
$m_1(k)$, $m_2(k)$	$k^{K_1^{(\mathrm{i})}}$, k^{1+q}	A.4
$m_3(q)$	$B_{k-1} > M = (\sqrt{2}-1)/(2\delta)$ and $k > M^{1/(1+q)}$	A.5
$m_4(y)$	$\min\{1, (\varepsilon+\kappa)^{K_1^{(\mathrm{i})}} + \sqrt{2}(1+q)y/\varepsilon\}$, $\kappa = M^{-1/(1+q)}$	A.6
$O(\cdot)$ in Lemma 4	$\lambda B_{k-1} = \sqrt{2}\,\delta B_{k-1}$	A.7
f_1	$\sqrt{2}\,\delta/(\cos\alpha - 1/\sqrt{2})$	A.8
f_2	$\max(f_{2a}, f_{2b})$, see step 3 of Section 3	A.8
f_3 in (11), (12)	$f_3 = f_{2a} + f_1/(2K_1^{(\mathrm{i})})$; (12) holds with $t\lambda B_{k-1}$	A.9
f_4, K_4	$f_3 + 3\delta$, $1/(\sqrt{2}(2+q))$	A.9
f_5	f_5' for $t \le k-1$; f_5'' for all $t \le k$	A.9
Lemma 7	$\beta_t \le \pi/8$, or $\beta_t \le \beta^* = \frac{1}{2}\arcsin q$	Lemma 4.1
K_5 in (18)	$1/(2\sqrt{2})$; in fact $1/(\sqrt{2}(t+1))$ in place of K_5/t	A.13
f_6, f_7, f_8 in (19)	$f_6 = f_5$; $f_7 = q^2 - \frac{1}{4}$; f_8, $f_8^{(k)}$ as in step 6	Prop. 4.2
K_6, K_7	$1/\mu$, $2/3 - \mu$	A.15
K_8, K_9, K_{10}, K_{11}	$K_7/\sqrt{2}$, $\sqrt{2}(1-\cos\beta_0)/K_7$, $1/(1+q)$, K_9/K_{10}	A.16
exponent $1/3$ in (29)–(32)	$\theta \in (0,1)$	A.17
m_5, K_{12}	$M/(2+q)$, $1/(2+q)$	A.17
K_{13}, K_{14}, K_{15}, f_9	f_9 as in step 8; trivial branch via $g_9 = \min(f_9, 1-\cos\beta_0)$	A.17
f_{10}	$g_9/3 + f_8^{(k)}$	A.18
f_{11}	$\frac{3}{2}g_9 + \frac{3}{2}\sqrt{2f_{10}} + f_5''$	A.20
f_{12}, f_{13}	$g_9/(1-g_9)$, $\sqrt{2g_9}/(1-g_9)$	A.21
f_{14}, f_{15}	$f_{11}/(1-g_9) + qf_{13}$, $\delta + qf_{12} + f_{14}$	A.22
c, f_{16}	$c = 1/2 - \gamma$ with free $\gamma > 0$ (Biró: $1/2 - c = f_{16} = \sqrt{f_{15}}$)	A.22
f_{17}	$m_4(f_{15}/\gamma)$	A.23
f_{18}, f_{19}	$\sqrt{f_7 + \gamma}$, $f_{18} + (q + \frac{1}{2})f_{17}$	A.24
f_{20}	$(2+2\sqrt{2})f_{19}$	A.25

gives $c_{n-1} = \sum_{j=0}^{n-1} b_j S_{n-j}$, so $|c_{n-1}| \le qB_{n-1}$ by (1). Since $B_{n-1} \ge 1$, this gives $q \ge 1/\sqrt{2}$, a contradiction. So $k := \min\{t : \alpha_t > \pi/4\}$ exists, $1 \le k \le n-1$, and (5), (6) hold.

A.3. **Lemma 2 (no constants).**

(i) For $1 \le t \le n-1$: $|c_{t-1} - tb_t| \le qB_{t-1}$.
Proof. By (3), $c_{t-1} - tb_t = \sum_{j=0}^{t-1} b_j S_{t-j}$. Each index $t-j$ lies in $[1,n]$, so $|S_{t-j}| \le q$ by (1). Use the triangle inequality.

(ii) For $1 \le t \le n$: $|c_{t-1}| \ge 1 + |b_1|\cos\alpha_1 + \cdots + |b_{t-1}|\cos\alpha_{t-1}$.
Proof. Induction on t. For $t = 1$, $|c_0| = 1$. Step $t \to t+1$ ($t \le n-1$): Lemma 1′(iv′) with $w_1 = c_{t-1}$, $w_2 = b_t$.

$$(\star) \qquad |c_{t-1}| \ge \Big(1 - \frac{1}{\sqrt{2}}\Big) + \frac{B_{t-1}}{\sqrt{2}} \ge \frac{B_{t-1}}{\sqrt{2}} \qquad (1 \le t \le k); \quad \text{in particular } c_{t-1} \ne 0.$$

Proof. In (ii), every $j \le t-1 \le k-1$ has $\alpha_j \le \pi/4$ by (5), so $\cos\alpha_j \ge 1/\sqrt{2}$. Hence $|c_{t-1}| \ge 1 + (B_{t-1} - 1)/\sqrt{2}$.

(iii) For $1 \le t \le n-1$: $|c_{t-1} - tb_t| \ge |c_{t-1}|\sin\alpha_t$.
Proof. Lemma 1′(iii′) with $w_1 = c_{t-1}$, $w_2 = tb_t$, since $a(c_{t-1}, tb_t) = \alpha_t$.

The angle α_k. $\pi/4 < \alpha_k < \pi/2$. Hence $b_k \neq 0$ and $\sin\alpha_k > 1/\sqrt{2}$.
Proof. $\alpha_k > \pi/4$ is (6). Suppose $\alpha_k \geq \pi/2$. We have $c_{k-1} \neq 0$ by $(\star)$. If $b_k = 0$, then $|c_{k-1} - kb_k| = |c_{k-1}|$. If $b_k \neq 0$, then $a(c_{k-1}, -kb_k) = \pi - \alpha_k \in [0, \pi/2]$, so Lemma 1′(iv′) gives $|c_{k-1} - kb_k| \geq |c_{k-1}| + k|b_k|\cos(\pi - \alpha_k) \geq |c_{k-1}|$. In both cases (i) (valid since $k \leq n-1$) and $(\star)$ give $B_{k-1}/\sqrt{2} \leq |c_{k-1}| \leq |c_{k-1} - kb_k| \leq qB_{k-1}$, so $q \geq 1/\sqrt{2}$, a contradiction. So $\alpha_k < \pi/2$; in particular $\alpha_k \neq \pi$, so $b_k \neq 0$. Since sin is increasing on $[0, \pi/2]$, $\sin\alpha_k > \sin(\pi/4) = 1/\sqrt{2}$.

(iv) $|c_{k-1}| < \sqrt{2}\, q\, B_{k-1}$.
Proof. (i) and (iii) at $t = k$ give $|c_{k-1}|\sin\alpha_k \leq qB_{k-1}$. Divide by $\sin\alpha_k > 1/\sqrt{2}$ (and $qB_{k-1} > 0$).

A.4. **Lemma 3.** Put

$$K_1^{(\mathrm{i})} := \frac{1}{\sqrt{2}} - q, \qquad K_2^{(\mathrm{i})} := 1 + q, \qquad K_1^{(\mathrm{ii})} := \frac{1}{\sqrt{2}(1+q)}, \qquad K_2^{(\mathrm{ii})} := 1 + \frac{1}{1/\sqrt{2} - q}.$$

All four are positive because $q < 1/\sqrt{2}$; moreover $0 < K_1^{(\mathrm{i})} < 1/\sqrt{2} - 1/2 < 1$ and $K_2^{(\mathrm{i})} > 1$.

(i) For $1 \leq t \leq k$: $K_1^{(\mathrm{i})} B_{t-1} \leq t|b_t| \leq K_2^{(\mathrm{i})} B_{t-1}$. In particular $b_t \neq 0$ for $1 \leq t \leq k$.
Proof. As $t \leq k \leq n-1$, Lemma 2(i) applies. Upper bound: $t|b_t| \leq |c_{t-1}| + |c_{t-1} - tb_t| \leq B_{t-1} + qB_{t-1}$, using $|c_{t-1}| \leq B_{t-1}$. Lower bound: $t|b_t| \geq |c_{t-1}| - |c_{t-1} - tb_t| \geq B_{t-1}/\sqrt{2} - qB_{t-1}$, by $(\star)$. Since $B_{t-1} \geq 1$ and $K_1^{(\mathrm{i})} > 0$, $b_t \neq 0$.

(ii) For $1 \leq t \leq k-1$: $K_1^{(\mathrm{ii})} \leq |c_t|/(t|b_t|) \leq K_2^{(\mathrm{ii})}$.
Proof. $b_t \neq 0$ by (i). Upper bound: $|c_t| \leq B_t = B_{t-1} + |b_t| \leq t|b_t|/K_1^{(\mathrm{i})} + |b_t|$, so $|c_t|/(t|b_t|) \leq 1/K_1^{(\mathrm{i})} + 1/t \leq 1 + 1/K_1^{(\mathrm{i})}$. Lower bound: $t+1 \leq k$, so $(\star)$ at index $t+1$ gives $|c_t| \geq B_t/\sqrt{2} \geq B_{t-1}/\sqrt{2} \geq t|b_t|/(\sqrt{2}\, K_2^{(\mathrm{i})})$.

(iii) For integers $1 \leq s \leq u \leq k$:

$$\Big(\frac{u}{s}\Big)^{K_1^{(\mathrm{i})}} \leq \frac{B_{u-1}}{B_{s-1}} \leq \Big(\frac{u}{s}\Big)^{K_2^{(\mathrm{i})}}.$$

Proof. For $1 \leq t \leq k$, $B_t/B_{t-1} = 1 + |b_t|/B_{t-1}$ lies in $[1 + K_1^{(\mathrm{i})}/t,\ 1 + K_2^{(\mathrm{i})}/t]$ by (i). Bernoulli's inequality says that for $x > -1$, $(1+x)^r \leq 1 + rx$ if $0 \leq r \leq 1$, and $(1+x)^r \geq 1 + rx$ if $r \geq 1$. With $x = 1/t$ and $r = K_1^{(\mathrm{i})} \in (0,1)$, resp. $r = K_2^{(\mathrm{i})} \geq 1$:

$$(1 + 1/t)^{K_1^{(\mathrm{i})}} \leq B_t/B_{t-1} \leq (1 + 1/t)^{K_2^{(\mathrm{i})}}.$$

Multiply over $t = s, \ldots, u-1$ (all $\leq k-1$); the product $\prod_{t=s}^{u-1}(1 + 1/t) = u/s$ telescopes. For $s = u$ all sides equal 1.

m_1, m_2. (iii) with $s = 1$, $u = k$ and $B_0 = 1$ gives $k^{K_1^{(\mathrm{i})}} \leq B_{k-1} \leq k^{1+q}$. So $m_1(k) := k^{K_1^{(\mathrm{i})}}$ and $m_2(k) := k^{1+q}$ work (S10). Biró asks $m_j(k) > 1$; ours equal 1 at $k = 1$, which does not matter because $k \geq 2$ (A.5).

(T) Trivial bounds. $|c_t| \leq B_t$ for every t, and $C_t \leq (t+1)B_t$; in particular $C_{k-1} \leq kB_{k-1}$. (Triangle inequality, and $|c_j| \leq B_j \leq B_t$ for $j \leq t$.)

Biró uses one pair of constants for (i) and (ii), namely $\min(K_1^{(\mathrm{i})}, K_1^{(\mathrm{ii})})$ and $\max(K_2^{(\mathrm{i})}, K_2^{(\mathrm{ii})})$. We keep the four constants separate (bookkeeping (B3)), because the large $K_2^{(\mathrm{ii})} \approx 5.8$ would weaken m_2 and Corollary 1.

A.5. **Corollary 1.** Let $M := (\sqrt{2} - 1)/(2\delta)$. Then $B_{k-1} > M$, $k > M^{1/(1+q)}$, and $M > 1$; hence $k \geq 2$.

Proof. $(\star)$ at $t = k$ gives $|c_{k-1}| \geq (1 - 1/\sqrt{2}) + B_{k-1}/\sqrt{2}$, and Lemma 2(iv) gives $|c_{k-1}| < \sqrt{2}\, qB_{k-1}$. So

$$1 - \frac{1}{\sqrt{2}} < \Big(\sqrt{2}\, q - \frac{1}{\sqrt{2}}\Big) B_{k-1} = \sqrt{2}\, \delta B_{k-1},$$

that is, $B_{k-1} > (1 - 1/\sqrt{2})/(\sqrt{2}\,\delta) = M$. By A.4, $k^{1+q} \geq B_{k-1} > M$, so $k > M^{1/(1+q)}$. Finally $M - 1 = (1/\sqrt{2} - q)/\delta > 0$. Since $B_0 = 1 < M < B_{k-1}$, we have $k \neq 1$. □

Biró's $m_3(q)$ can be taken as $M^{1/(1+q)}$; it satisfies $m_3 < k$ and $m_3 \le M < B_{k-1}$.

A.6. **Corollary 2.** Let $\kappa := M^{-1/(1+q)}$. By A.5, $1/k < \kappa$.

Corollary 2 of [Bir00b], explicit form. *Let $H \subseteq \{0,1,\dots,k-1\}$ and $y \ge 0$, and suppose*

$$\sum_{t\in H} |c_t| \le y\, C_{k-1}. \tag{7}$$

Then for every $\varepsilon \in (0,1)$

$$\sum_{t\in H} |b_t| \le \Big[(\varepsilon + 1/k)^{K_1^{(\mathrm{i})}} + \frac{y}{\varepsilon K_1^{(\mathrm{ii})}}\Big] B_{k-1}. \tag{8$'$}$$

Hence $\sum_{t\in H}|b_t| \le m_4(y)\, B_{k-1}$ with

$$m_4(y) := \min\Big\{1,\ (\varepsilon+\kappa)^{1/\sqrt{2}-q} + \sqrt{2}(1+q)\,\frac{y}{\varepsilon}\Big\}.$$

Proof. Let $s := \lfloor \varepsilon k \rfloor$. Since $0 < \varepsilon k < k$, we have $0 \le s \le k-1$. Split $H = H_1 \cup H_2$ with $H_1 = H \cap \{0,\dots,s\}$ and $H_2 = H \cap \{s+1,\dots,k-1\}$.

(a) $\sum_{t\in H_1}|b_t| \le \sum_{t=0}^{s}|b_t| = B_s$ (recall $b_0 = 1$). Lemma 3(iii) with the pair $(s+1,k)$ gives $B_{k-1}/B_s \ge (k/(s+1))^{K_1^{(\mathrm{i})}}$. So $B_s \le ((s+1)/k)^{K_1^{(\mathrm{i})}} B_{k-1} \le (\varepsilon+1/k)^{K_1^{(\mathrm{i})}} B_{k-1}$, because $s+1 \le \varepsilon k+1$ and $K_1^{(\mathrm{i})} > 0$.

(b) For $t \in H_2$ we have $1 \le t \le k-1$ and $t \ge s+1 > \varepsilon k$. Lemma 3(ii) gives $t|b_t| \le |c_t|/K_1^{(\mathrm{ii})}$, so $|b_t| \le |c_t|/(K_1^{(\mathrm{ii})}\varepsilon k)$. Hence, by (7) and (T),

$$\sum_{t\in H_2}|b_t| \le \frac{1}{K_1^{(\mathrm{ii})}\varepsilon k}\sum_{t\in H}|c_t| \le \frac{yC_{k-1}}{K_1^{(\mathrm{ii})}\varepsilon k} \le \frac{yB_{k-1}}{K_1^{(\mathrm{ii})}\varepsilon}.$$

Adding (a) and (b) gives (8′). For m_4: $1/k < \kappa$, the function $x \mapsto (\varepsilon+x)^{K_1^{(\mathrm{i})}}$ is increasing, $1/K_1^{(\mathrm{ii})} = \sqrt{2}(1+q)$, and trivially $\sum_{t\in H}|b_t| \le \sum_{t=0}^{k-1}|b_t| = B_{k-1}$. □

Biró argues that if y is small and (7) holds with some nonempty H, then k is large, so that k may be assumed arbitrarily large. We use Corollary 1 for the size of k instead (bookkeeping (B4)); our form needs no smallness assumption on y.

A.7. **Lemma 4.** For $1 \le t \le k$:

$$\frac{B_{t-1}}{\sqrt{2}} \le |c_{t-1}| \le \frac{B_{t-1}}{\sqrt{2}} + \lambda B_{k-1}, \qquad \lambda = \sqrt{2}\,\delta.$$

Proof. The lower bound is (⋆). For $t \le j \le k-1$, Lemma 1′(iv′) with $w_1 = c_{j-1}$, $w_2 = b_j$ and $\cos\alpha_j \ge 1/\sqrt{2}$ (by (5)) gives $|c_j| \ge |c_{j-1}| + |b_j|/\sqrt{2}$. Adding over $j = t,\dots,k-1$ (an empty sum if $t = k$) gives $|c_{k-1}| \ge |c_{t-1}| + (B_{k-1} - B_{t-1})/\sqrt{2}$. With Lemma 2(iv),

$$|c_{t-1}| \le \sqrt{2}\,qB_{k-1} - \frac{B_{k-1}-B_{t-1}}{\sqrt{2}} = \frac{B_{t-1}}{\sqrt{2}} + \sqrt{2}\,\delta B_{k-1}.$$ □

A.8. **Lemma 5.** Let $H_\alpha := \{1 \le t \le k-1 : \alpha_t \le \alpha\}$.

(i) $\sum_{t\in H_\alpha}|b_t| \le f_1 B_{k-1}$, where $f_1 := \sqrt{2}\,\delta/(\cos\alpha - 1/\sqrt{2})$.

Proof. Use Lemma 2(ii) at $t = k$. For $j \in H_\alpha$, $\cos\alpha_j \ge \cos\alpha$; for $1 \le j \le k-1$ with $j \notin H_\alpha$, $\cos\alpha_j \ge 1/\sqrt{2}$ by (5). With $X := \sum_{j\in H_\alpha}|b_j|$,

$$|c_{k-1}| \ge 1 + \frac{B_{k-1}-1-X}{\sqrt{2}} + X\cos\alpha = 1 + \frac{B_{k-1}-1}{\sqrt{2}} + \Big(\cos\alpha - \frac{1}{\sqrt{2}}\Big)X \ge \frac{B_{k-1}}{\sqrt{2}} + \Big(\cos\alpha - \frac{1}{\sqrt{2}}\Big)X.$$

With Lemma 2(iv), $(\cos\alpha - 1/\sqrt{2})X \le (\sqrt{2}q - 1/\sqrt{2})B_{k-1} = \sqrt{2}\,\delta B_{k-1}$. Divide by $\cos\alpha - 1/\sqrt{2} > 0$.

(ii-b) For $1 \le t \le k-1$ with $t \notin H_\alpha$: $0 \le |c_{t-1}| - B_{t-1}/\sqrt{2} \le f_{2b}B_{t-1}$, where $f_{2b} := q/\sin\alpha - 1/\sqrt{2} > 0$.

Proof. $t \notin H_\alpha$ and (5) give $\alpha < \alpha_t \le \pi/4$, so $\sin\alpha_t > \sin\alpha > 0$. Lemma 2(iii) and (i) give

$|c_{t-1}|\sin\alpha_t \le qB_{t-1}$, so $|c_{t-1}| \le (q/\sin\alpha)B_{t-1}$ (compare Biró's (10)). The lower bound is $(\star)$. Positivity: $q/\sin\alpha - 1/\sqrt{2} = (q - \sin\alpha/\sqrt{2})/\sin\alpha$, and $\sin\alpha/\sqrt{2} < 1/2 < q$.

(ii-a) For $1 \le t \le k-1$ with $t \notin H_\alpha$: $|t|b_t| - B_{t-1}/2| \le f_{2a}B_{t-1}$, where

$$f_{2a} := \sqrt{q^2 - \tfrac{1}{2}\sin^2\alpha} + \min\Big(1, \frac{q}{\sin\alpha}\Big)\Big(\cos\alpha - \frac{1}{\sqrt{2}}\Big) + \frac{f_{2b}}{\sqrt{2}}.$$

Proof. Write $c := c_{t-1}$ and $B := B_{t-1}$. We have $c \neq 0$ by $(\star)$ and $b_t \neq 0$ by Lemma 3(i), and the angle between c and tb_t has absolute value α_t. Lemma 1(i) of [Bir00b] with $w_1 = c$, $w_2 = tb_t$, together with Lemma 2(i), gives (Biró's (9))

$$q^2B^2 \ge |c - tb_t|^2 = |c|^2\sin^2\alpha_t + (t|b_t| - |c|\cos\alpha_t)^2.$$

By $(\star)$, $|c|^2 \ge B^2/2$, and $\sin^2\alpha_t > \sin^2\alpha$. So

$$(t|b_t| - |c|\cos\alpha_t)^2 \le B^2(q^2 - \tfrac{1}{2}\sin^2\alpha_t) \le B^2(q^2 - \tfrac{1}{2}\sin^2\alpha),$$

where $q^2 - \frac{1}{2}\sin^2\alpha > \frac{1}{4} - \frac{1}{4} = 0$. Hence $|t|b_t| - |c|\cos\alpha_t| \le B\sqrt{q^2 - \frac{1}{2}\sin^2\alpha}$. Now use the identity

$$t|b_t| - \frac{B}{2} = (t|b_t| - |c|\cos\alpha_t) + |c|\Big(\cos\alpha_t - \frac{1}{\sqrt{2}}\Big) + \frac{1}{\sqrt{2}}\Big(|c| - \frac{B}{\sqrt{2}}\Big).$$

In the second term, $0 \le \cos\alpha_t - 1/\sqrt{2} \le \cos\alpha - 1/\sqrt{2}$ (since $\alpha < \alpha_t \le \pi/4$), and $|c| \le \min(1, q/\sin\alpha)B$ (by (T) and the proof of (ii-b)). The third term is at most $f_{2b}B/\sqrt{2}$ in absolute value, by (ii-b). The triangle inequality gives the claim.

Biró's $f_2 := \max(f_{2a}, f_{2b})$ satisfies both inequalities of his Lemma 5(ii). As $\alpha \to \pi/4$ and then $q \to 1/2$, $f_1, f_{2a}, f_{2b} \to 0$, as Biró requires.

A.9. **Lemma 6.** Biró: for $1 \le t \le k$, $t|c_t|/C_{t-1} = 3/2 + O(f_5B_{k-1}/B_t)$. For $t \ge 1$ put $P_t := \sum_{j=1}^t j|b_j|$ and $A_t := \sum_{j=1}^t B_{j-1}$; note $C_{t-1} = \sum_{j=1}^t |c_{j-1}|$. Define

$$f_3 := f_{2a} + \frac{f_1}{2K_1^{(\mathrm{i})}}, \quad f_4 := f_3 + 3\delta, \quad K_4 := \frac{1}{\sqrt{2}(2+q)}, \quad f_5' := (2+q)(f_3 + 3\delta),$$

$$f_5'' := (2+q)\Big(f_3 + 3\delta + (\sqrt{2}(1+q) + \tfrac{1}{2})\kappa\Big) \quad (\ge f_5').$$

(13) For every $t \ge 1$: $P_t + A_t = tB_t$.

Proof. $tB_t = \sum_{j=1}^t (jB_j - (j-1)B_{j-1}) = \sum_{j=1}^t (j(B_j - B_{j-1}) + B_{j-1}) = \sum_{j=1}^t (j|b_j| + B_{j-1})$.

(11) For $1 \le t \le k-1$: $|P_t - A_t/2| \le tf_3B_{k-1}$.

Proof. $P_t - A_t/2 = \sum_{j=1}^t (j|b_j| - B_{j-1}/2)$, and every j here satisfies $j \le k-1$. For $j \notin H_\alpha$, Lemma 5(ii-a) gives $|j|b_j| - B_{j-1}/2| \le f_{2a}B_{j-1} \le f_{2a}B_{k-1}$; there are at most t such j. For $j \in H_\alpha$, both $j|b_j|$ and $B_{j-1}/2$ are ≥ 0, so $|j|b_j| - B_{j-1}/2| \le \max(j|b_j|, B_{j-1}/2)$. By Lemma 3(i), $B_{j-1}/2 \le j|b_j|/(2K_1^{(\mathrm{i})})$, and $1/(2K_1^{(\mathrm{i})}) > 1$ because $K_1^{(\mathrm{i})} < 1/2$. So the term is at most $j|b_j|/(2K_1^{(\mathrm{i})}) \le t|b_j|/(2K_1^{(\mathrm{i})})$. Adding over $j \in H_\alpha$ and using Lemma 5(i) gives at most $tf_1B_{k-1}/(2K_1^{(\mathrm{i})})$.

(12′) For $1 \le t \le k$: $0 \le C_{t-1} - A_t/\sqrt{2} \le t\lambda B_{k-1}$.

Proof. Add Lemma 4 over the indices $1, \dots, t$ (all $\le k$): $0 \le |c_{j-1}| - B_{j-1}/\sqrt{2} \le \lambda B_{k-1}$. (Biró obtains (12) from Lemma 5; this is S9.)

Biró's f_4. For $1 \le t \le k-1$: $\big|\frac{3}{\sqrt{2}}C_{t-1} - tB_t\big| \le tf_4B_{k-1}$.

Proof. Put $E_A := A_t - \frac{2}{3}tB_t$ and $E_c := C_{t-1} - A_t/\sqrt{2}$. By (13), $tB_t = \frac{3}{2}A_t + (P_t - A_t/2)$, so $E_A = -\frac{2}{3}(P_t - A_t/2)$ and $|E_A| \le \frac{2}{3}tf_3B_{k-1}$ by (11). By (12′), $0 \le E_c \le t\lambda B_{k-1}$. The identity $\frac{3}{\sqrt{2}}C_{t-1} - tB_t = \frac{3}{2}E_A + \frac{3}{\sqrt{2}}E_c$ gives the bound $tf_3B_{k-1} + \frac{3}{\sqrt{2}}t\lambda B_{k-1} = t(f_3 + 3\delta)B_{k-1}$.

K_4. For $1 \le t \le k$: $C_{t-1} \ge K_4tB_t$.

Proof. By Lemma 3(i), $j|b_j| \le K_2^{(\mathrm{i})}B_{j-1}$ for $j \le t \le k$, so $P_t \le K_2^{(\mathrm{i})}A_t$, and (13) gives $tB_t \le (1 + K_2^{(\mathrm{i})})A_t$. By (12′), $C_{t-1} \ge A_t/\sqrt{2} \ge tB_t/(\sqrt{2}(2+q))$.

Lemma 6 for $1 \le t \le k-1$: $|t|c_t|/C_{t-1} - 3/2| \le f_5' B_{k-1}/B_t$.

Proof. Lemma 4 at index $t+1 \le k$ gives $|c_t| = B_t/\sqrt{2} + e_1$ with $0 \le e_1 \le \lambda B_{k-1}$. With E_A, E_c as above, one has the identity

$$t|c_t| - \tfrac{3}{2}C_{t-1} = (te_1 - \tfrac{3}{2}E_c) - \tfrac{3}{2\sqrt{2}}E_A.$$

Since $te_1 \in [0, t\lambda B_{k-1}]$ and $\frac{3}{2}E_c \in [0, \frac{3}{2}t\lambda B_{k-1}]$, we get $|te_1 - \frac{3}{2}E_c| \le \frac{3}{2}t\lambda B_{k-1}$. Also $|\frac{3}{2\sqrt{2}}E_A| \le \frac{1}{\sqrt{2}}tf_3 B_{k-1}$. Hence

$$|t|c_t| - \tfrac{3}{2}C_{t-1}| \le tB_{k-1}(\tfrac{3}{2}\lambda + f_3/\sqrt{2}) = tB_{k-1}\,\frac{3\delta + f_3}{\sqrt{2}}.$$

Divide by $C_{t-1} \ge K_4 t B_t$ and use $1/(\sqrt{2}K_4) = 2 + q$.

Lemma 6 for $t = k$: $|k|c_k|/C_{k-1} - 3/2| \le f_5'' B_{k-1}/B_k$.

Proof. Write $B := B_{k-1}$, and recall $k \ge 2$, so $t = k-1 \ge 1$ is a valid index in (11) and (13).

- Lemma 4 at $t = k$: $|c_{k-1}| = B/\sqrt{2} + e$ with $0 \le e \le \lambda B$.
- $c_k = c_{k-1} + b_k$. Put $r := k(|c_k| - |c_{k-1}|)$. Then $|r| \le k|b_k| \le K_2^{(\mathrm{i})} B$ (Lemma 3(i) at $t = k$), and $k|c_k| = kB/\sqrt{2} + ke + r$.
- Put $E_A := A_{k-1} - \frac{2}{3}(k-1)B$. As in the proof of f_4 (with $t = k-1$), $|E_A| \le \frac{2}{3}(k-1)f_3 B$. Then $A_k = A_{k-1} + B = \frac{2}{3}kB + \frac{1}{3}B + E_A$.
- Put $E_c := C_{k-1} - A_k/\sqrt{2}$. By (12′) at $t = k$, $0 \le E_c \le k\lambda B$.

One has the identity

$$k|c_k| - \tfrac{3}{2}C_{k-1} = (ke - \tfrac{3}{2}E_c) + r - \frac{B}{2\sqrt{2}} - \tfrac{3}{2\sqrt{2}}E_A.$$

As before, $|ke - \frac{3}{2}E_c| \le \frac{3}{2}k\lambda B$. So

$$|k|c_k| - \tfrac{3}{2}C_{k-1}| \le B\Big(\tfrac{3}{2}k\lambda + K_2^{(\mathrm{i})} + \frac{1}{2\sqrt{2}} + \frac{(k-1)f_3}{\sqrt{2}}\Big) \le kB\Big(\frac{3\delta + f_3}{\sqrt{2}} + \frac{K_2^{(\mathrm{i})} + 1/(2\sqrt{2})}{k}\Big).$$

Divide by $C_{k-1} \ge K_4 k B_k$ (K_4 at $t = k$); then use $1/(\sqrt{2}K_4) = 2 + q$, $\sqrt{2}(K_2^{(\mathrm{i})} + 1/(2\sqrt{2})) = \sqrt{2}(1+q) + \frac{1}{2}$, and $1/k < \kappa$.

So Biró's Lemma 6 holds for all $1 \le t \le k$ with $f_5 := f_5''$. The extra term $(\sqrt{2}(1+q) + \frac{1}{2})\kappa$ is the cost of $t = k$, which Biró treats in one sentence ("since $|c_k| = |c_{k-1}| + O(|b_k|)$").

A.10. **Formula (14).** For $1 \le t \le n-1$, (14) holds, and hence

$$|2D_{t-1} - tc_t| \le qC_{t-1} \qquad (1 \le t \le n-1). \tag{14′}$$

Proof. Let $L_t := \sum_{j=0}^{t-1} c_j S_{t-j}$ and $R_t := 2D_{t-1} - tc_t$. For $t = 1$: $L_1 = S_1$ and $R_1 = 2 - c_1 = 1 - b_1$, which are equal by (3) at $t = 1$. For $2 \le t \le n-1$, reindexing gives $L_t - L_{t-1} = S_t + \sum_{j=1}^{t-1}(c_j - c_{j-1})S_{t-j} = \sum_{j=0}^{t-1} b_j S_{t-j}$, which is $c_{t-1} - tb_t$ by (3). Also $R_t - R_{t-1} = 2c_{t-1} - tc_t + (t-1)c_{t-1} = (t+1)c_{t-1} - t(c_{t-1} + b_t) = c_{t-1} - tb_t$. Induction gives $L_t = R_t$. Then (14′) follows from (1) and the triangle inequality. □

(Biró states (14) for $t \le k$; it holds for $t \le n-1$, and $k \le n-1$.)

A.11. **Lemma 7 and (17).** Lemma 7 of [Bir00b], in the general form we use, is Lemma 4.1. Let $\beta_0 \in \{\pi/8, \beta^*\}$; then $\beta_t \le \beta_0$ for $1 \le t \le k$, by Lemma 4.1(a) with $\beta = \pi/8$ (since $\sin(\pi/4) = 1/\sqrt{2} > q$), resp. by Lemma 4.1(b). Put $h_t := |D_{t-1}|/C_{t-1}$.

$$\cos\beta_0 \le h_t \le 1 \qquad (1 \le t \le k); \qquad \text{in particular } D_{t-1} \ne 0. \tag{17}$$

Proof. The upper bound is the triangle inequality. The lower bound is the induction of (15) in the proof of Lemma 4.1, with β_0 in place of β; it is valid because $\beta_j \le \beta_0$ for all $j \le t-1 \le k-1$. (The same argument gives (17) also for $t = k+1$.)

A.12. **Lemma 8: notation and statement.** Biró's Lemma 8 states (i) $|D_{k-1}|/C_{k-1} = 1+O(f_9)$ and (ii) $\cos\beta_k = 1 + O(f_{10})$. Notation:

- h_t for $1 \le t \le k+1$ as above; $X_t := B_{k-1}/B_t$ for $1 \le t \le k$. Since B is nondecreasing, $X_t \ge 1$ for $t \le k-1$ and $X_k \le 1$.
- $F(h) := (2h^2+1)/(3h)$ for $h > 0$, with fixed point 1.
- $\rho_t := t|c_t|/C_{t-1} - 3/2$ for $1 \le t \le k$. By Lemma 6, $|\rho_t| \le f_5' X_t$ for $t \le k-1$ and $|\rho_k| \le f_5'' X_k$.
- Since $\beta_0 \le \pi/8 < \pi/4$, $\cos\beta_0 > 1/\sqrt{2}$, so $\cos^2\beta_0 > 1/2$.

Free parameters: $\theta \in (0,1)$ and $\mu \in (0, 2/3)$. Constants:

$$f_7 := q^2 - \tfrac14, \quad f_8 := \frac{f_7 + f_5'}{6\cos\beta_0}, \quad f_8^{(k)} := \frac{f_7 + f_5''}{6\cos\beta_0}, \quad K_6 := \frac{1}{\mu}, \quad K_7 := \frac{2}{3} - \mu,$$

$$K_8 := \frac{K_7}{\sqrt{2}}, \quad K_9 := \frac{\sqrt{2}(1-\cos\beta_0)}{K_7}, \quad K_{10} := \frac{1}{1+q}, \quad K_{11} := \frac{K_9}{K_{10}} = \frac{\sqrt{2}(1+q)(1-\cos\beta_0)}{K_7},$$

$$m_5 := \frac{M}{2+q}, \quad K_{12} := \frac{1}{2+q}, \quad f_9 := \max\Big\{\frac{f_8^{1-\theta}}{\mu}, \frac{K_{11}}{\log m_5}, \frac{K_{11}}{\log(K_{12} f_8^{-\theta})}\Big\},$$

$$f_{10} := \frac{\min(f_9, 1-\cos\beta_0)}{3} + f_8^{(k)}.$$

Lemma 8 of [Bir00b], explicit form. *Suppose* (L8a) $m_5 > 1$ *and* (L8b) $(2+q)f_8^\theta < 1$. *Then* (i) $1 - f_9 \le h_k \le 1$, *and* (ii) $1 - f_{10} \le \cos\beta_k \le 1$.

Without (L8a), (L8b) we still have $h_k \ge \cos\beta_0$ by (17), and (ii) holds with f_9 replaced by $1-\cos\beta_0$ (A.18). Note that (L8b) implies $f_8 < 1$. The proof takes A.13–A.18.

A.13. **One step of the recursion: (18).** For $1 \le t \le k$ we have $\beta_t \le \beta_0 < \pi$, so by the convention for $a(\cdot,\cdot)$, $D_{t-1} \ne 0$, $c_t \ne 0$, $\beta_t = |\arg(c_t/D_{t-1})|$, and $\cos\beta_t \ge \cos\beta_0 > 0$.

(18a) For $1 \le t \le k$:

$$h_{t+1} \ge \frac{h_t C_{t-1} + |c_t|\cos\beta_t}{C_t} = h_t + (\cos\beta_t - h_t)\frac{|c_t|}{C_t}.$$

Proof. $D_t = D_{t-1} + c_t$. Lemma 1′(iv′) with $w_1 = D_{t-1}$, $w_2 = c_t$ gives $|D_t| \ge |D_{t-1}| + |c_t|\cos\beta_t = h_t C_{t-1} + |c_t|\cos\beta_t$. Divide by $C_t = C_{t-1} + |c_t| > 0$. The equality is a rearrangement.

(18b) For $1 \le t \le k-1$: $|c_t|/C_t \ge 1/(\sqrt{2}(t+1))$ (S5): $(\star)$ at index $t+1 \le k$ gives $|c_t| \ge B_t/\sqrt{2}$, and $C_t \le (t+1)B_t$ by (T).

(18′) For $1 \le t \le k-1$, if $\cos\beta_t \ge h_t$, then $h_{t+1} \ge h_t + (\cos\beta_t - h_t)/(\sqrt{2}(t+1))$. (By (18a) and (18b).)

(18c) For $1 \le t \le k$ and real H: if $h_t \ge H$ and $\cos\beta_t \ge H$, then $h_{t+1} \ge H$.

Proof. By (18a), $h_{t+1} - H \ge \big((h_t - H)C_{t-1} + (\cos\beta_t - H)|c_t|\big)/C_t \ge 0$.

Biró's (18) is printed as $h_{t-1} \ge h_t + K_5(\cos\beta_t - h_t)/t$; the index should be $t+1$ (§4.1). His form with $1/t$ holds with $K_5 = 1/(2\sqrt{2})$, since $1/(\sqrt{2}(t+1)) \ge 1/(2\sqrt{2}\,t)$ for $t \ge 1$.

A.14. **The inequality (19).** This is Proposition 4.2: for $1 \le t \le k-1$, $\cos\beta_t \ge F(h_t) - f_8 X_t$, and $\cos\beta_k \ge F(h_k) - f_8^{(k)}$. In Biró's form $O(f\,X_k^2)$ at $t = k$, note that $X_k \le 1$, so $X_k^2 \le X_k$; his form still holds there with a changed constant, because $X_k \ge 1/(1 + K_2^{(\mathrm{i})}/k) \ge 1/(2+q)$ by Lemma 3(i). Our (19′) at $t = k$ uses $X_k \le 1$ directly.

A.15. **The two cases: (20)–(26).** *(20′)* $F'(h) = (2h^2-1)/(3h^2)$, so F is increasing on $[1/\sqrt{2}, \infty)$. $(F(h) - h)' = -(1+h^2)/(3h^2) < 0$, so $F(h) - h = (1-h^2)/(3h)$ is decreasing on $(0,\infty)$. For $0 < H \le 1$, $F(H) - H \ge 2(1-H)/3$ (S3).

Lemma 8a (cases (25), (26)). *Let* $0 < \varphi < 1 - \cos\beta_0$ *and* $H := 1 - \varphi$, *so* $\cos\beta_0 < H < 1$. *Let* $1 \le t_0 \le k-1$ *and* $Q := B_{k-1}/B_{t_0}$, *and assume*

$$f_8 Q \le \mu\varphi. \tag{24′}$$

Then for every t *with* $t_0 \le t \le k-1$:

(25′) *if* $h_t \ge H$, *then* $\cos\beta_t \ge 1 - \varphi/3 - \mu\varphi \ge 1 - \varphi = H$;

(26′) *if* $h_t \le H$, *then* $\cos\beta_t - h_t \ge K_7\varphi$, *with* $K_7 = 2/3 - \mu > 0$.

Proof. For $t_0 \le t \le k-1$ we have $B_t \ge B_{t_0}$, so $X_t \le Q$, and (19′) with (24′) gives $\cos\beta_t \ge F(h_t) - f_8 Q \ge F(h_t) - \mu\varphi$. (25′): $h_t \in [H, 1]$ (as $h_t \le 1$ by (17)), and $H > \cos\beta_0 > 1/\sqrt{2}$. As F is increasing there, $F(h_t) \ge F(H) \ge H + 2\varphi/3 = 1 - \varphi/3$. Hence $\cos\beta_t \ge 1 - \varphi/3 - \mu\varphi$, and $(1 - \varphi/3 - \mu\varphi) - (1 - \varphi) = \varphi(2/3 - \mu) > 0$. (26′): $h_t \in [\cos\beta_0, H] \subset (0, 1]$. As $F(h) - h$ is decreasing, $F(h_t) - h_t \ge F(H) - H \ge 2\varphi/3$. Hence $\cos\beta_t - h_t \ge 2\varphi/3 - \mu\varphi = K_7\varphi$. □

Biró needs (24) "with a large enough constant K_6" and gets (26) with some K_7; ours are $K_6 = 1/\mu$ and $K_7 = 2/3 - \mu$. Biró's (25) also covers the case $1 - f_9 \le 1/\sqrt{2}$ via Lemma 7; we avoid that case by the trivial branch in A.17.

A.16. **(27)–(28): if h stays low, Q is small.**

Lemma 8b. *Under the assumptions of Lemma 8a, suppose* $h_t < H$ *for every* $t_0 \le t \le k-1$. *Then*

$$\varphi \log Q \le K_{11}. \tag{28′}$$

Proof. (27′) For $t_0 \le t \le k-1$, (26′) gives $\cos\beta_t - h_t \ge K_7\varphi > 0$. So (18′) applies, and $h_{t+1} - h_t \ge K_7\varphi/(\sqrt{2}(t+1)) = K_8\varphi/(t+1)$. Adding over $t = t_0, \dots, k-1$,

$$h_k - h_{t_0} \ge K_8\varphi \sum_{s=t_0+1}^{k} \frac{1}{s}.$$

By (17), $h_k \le 1$ and $h_{t_0} \ge \cos\beta_0$ (S4), so the left side is at most $1 - \cos\beta_0$. Hence $\varphi \sum_{s=t_0+1}^{k} 1/s \le (1 - \cos\beta_0)/K_8 = K_9$.

Harmonic sum against $\log Q$: for $1 \le s \le k$, Lemma 3(i) gives $B_s = B_{s-1} + |b_s| \le B_{s-1}(1 + K_2^{(\mathrm{i})}/s)$. Multiplying over $s = t_0 + 1, \dots, k-1$ (an empty product if $t_0 = k-1$, and then $Q = 1$),

$$Q = \frac{B_{k-1}}{B_{t_0}} \le \prod_{s=t_0+1}^{k-1} \Big(1 + \frac{K_2^{(\mathrm{i})}}{s}\Big).$$

Since $\log(1+u) \le u$ for $u > -1$, $\log Q \le K_2^{(\mathrm{i})} \sum_{s=t_0+1}^{k-1} 1/s \le K_2^{(\mathrm{i})} \sum_{s=t_0+1}^{k} 1/s$. So $\sum_{s=t_0+1}^{k} 1/s \ge K_{10} \log Q$ with $K_{10} = 1/K_2^{(\mathrm{i})} = 1/(1+q)$. Combining, $\varphi \log Q \le K_9/K_{10} = K_{11}$. □

Biró uses $h_{t_0} \ge 0$ in (27); we use $h_{t_0} \ge \cos\beta_0$ (S4), which makes K_9 smaller by the factor $1 - \cos\beta_0$. His harmonic sum runs over $t_0, \dots, k-1$ with weights K_5/t; ours over $t_0 + 1, \dots, k$ with weights $1/(\sqrt{2}(t+1))$ (S5).

A.17. **Choice of t_0 and proof of Lemma 8(i).** Assume (L8a) and (L8b), and put $\varphi := f_9$. Then $f_8 < 1$. We show $h_k \ge 1 - \varphi$; the bound $h_k \le 1$ is the triangle inequality.

Trivial branch. If $\varphi \ge 1 - \cos\beta_0$, then (17) gives $h_k \ge \cos\beta_0 \ge 1 - \varphi$, and we are done. (This replaces Biró's K_{14}, K_{15}.) So assume from now on that $0 < \varphi < 1 - \cos\beta_0$, and put $H := 1 - \varphi$. Recall $k \ge 2$.

Choice of t_0 (Biró's (29)–(32), with exponent θ instead of 1/3; S6):

- Case (29′): $f_8^\theta B_{k-1} \le 1$. Put $t_0 := 1$ (allowed, since $1 \le k-1$).
- Case (30′): $f_8^\theta B_{k-1} > 1$. Let t_0 be the least $t \in \{1, \dots, k-1\}$ with $f_8^\theta B_{k-1}/B_t < 1$. It exists, because at $t = k-1$ the left side is $f_8^\theta < 1$.

In both cases $1 \le t_0 \le k-1$. Put $Q := B_{k-1}/B_{t_0}$ as in Lemma 8a.

Claim 1: (24′) holds. We show $f_8 Q \le f_8^{1-\theta}$; then (24′) follows, since $\varphi \ge f_8^{1-\theta}/\mu$ by the definition of f_9. In case (29′), $Q = B_{k-1}/B_1 \le B_{k-1} \le f_8^{-\theta}$. In case (30′), $Q < f_8^{-\theta}$ by the choice of t_0.

Claim 2: $\varphi \log Q > K_{11}$.

- Case (29′): Lemma 3(i) at $t = 1$ gives $B_1 = 1 + |b_1| \le 1 + K_2^{(\mathrm{i})} = 2 + q$, and Corollary 1 gives $B_{k-1} > M$. So $Q > M/(2+q) = m_5 > 1$ by (L8a). Since $\varphi \ge K_{11}/\log m_5 > 0$, we get $\varphi \log Q > \varphi \log m_5 \ge K_{11}$.

- Case (30′): first, $f_8^{\theta} B_{k-1}/B_{t_0-1} \geq 1$ (if $t_0 = 1$, this is (30′) itself, with $B_0 = 1$, and it is strict; if $t_0 \geq 2$, it holds by the minimality of t_0). Second, Lemma 3(i) at $t = t_0$ gives $B_{t_0} \leq (1 + K_2^{(\mathrm{i})}/t_0) B_{t_0-1} \leq (2+q) B_{t_0-1}$, and the last inequality is strict when $t_0 \geq 2$. In both subcases one of the two inequalities is strict, so

$$Q = \frac{B_{k-1}}{B_{t_0-1}} \cdot \frac{B_{t_0-1}}{B_{t_0}} > \frac{f_8^{-\theta}}{2+q} = K_{12} f_8^{-\theta} > 1$$

by (L8b). Since $\varphi \geq K_{11}/\log(K_{12} f_8^{-\theta}) > 0$, we get $\varphi \log Q > \varphi \log(K_{12} f_8^{-\theta}) \geq K_{11}$.

Conclusion. If $h_t < H$ for every $t_0 \leq t \leq k-1$, Lemma 8b (which applies by Claim 1) gives $\varphi \log Q \leq K_{11}$, contradicting Claim 2. So $h_{t_1} \geq H$ for some t_1 with $t_0 \leq t_1 \leq k-1$. Then (25′) gives $\cos \beta_{t_1} \geq H$, and (18c) gives $h_{t_1+1} \geq H$. If $t_1 + 1 \leq k-1$, repeat with $t_1 + 1$ in place of t_1. After finitely many steps, $h_k \geq H = 1 - f_9$.

Biró defines $f_9 = K_{13} \max(1/\log(1/f_8), 1/\log m_5)$ with K_{13} "large enough", and treats $f_8 \geq K_{14}$ by a trivial branch with $f_9 \geq K_{15}$. At one fixed point (q, α) these are not needed: our f_9 is the three-term maximum, and its third term is about $K_{11}/(\theta \log(1/f_8))$ (float), which matches his $K_{13}/\log(1/f_8)$. His m_5 is ours, and his $K_{12} = 1/(1+K_2)$ is ours with $K_2 = K_2^{(\mathrm{i})}$. Because (19′) is linear in Q, any $\theta \in (0,1)$ works; with the quadratic form of (19) one needs $\theta < 1/2$.

A.18. **Lemma 8(ii).** Let $\varphi' := \min(f_9, 1 - \cos\beta_0)$ and $H' := 1 - \varphi'$. By Lemma 8(i) and (17), $H' \leq h_k \leq 1$, and $H' \geq \cos\beta_0 > 1/\sqrt{2}$. As F is increasing on $[1/\sqrt{2}, 1]$, (20′) gives

$$F(h_k) \geq F(H') \geq H' + \tfrac{2}{3}(1 - H') = 1 - \varphi'/3.$$

With (19′) at $t = k$: $\cos\beta_k \geq 1 - \varphi'/3 - f_8^{(k)} = 1 - f_{10}$; and $\cos\beta_k \leq 1$. Without (L8a), (L8b) the same proof works with $\varphi' = 1 - \cos\beta_0$, which (17) gives for free. (Biró: "the case $t = k$ of (19) shows that (ii) is a consequence of (i)".)

A.19. **Section 4 of [Bir00b]: notation and inputs.** Biró's Section 4 turns Lemma 8 into a contradiction in steps (38)–(47). Recall $2 \leq k \leq n-1$. Write

$$B := B_{k-1}, \quad C := C_{k-1}, \quad D := D_{k-1}, \quad h := h_k = |D|/C.$$

Put $g_9 := \min(f_9, 1 - \cos\beta_0)$ if (L8a) and (L8b) hold, and $g_9 := 1 - \cos\beta_0$ otherwise. Then $0 < g_9 \leq 1 - \cos\beta_0 \leq 1 - \cos(\pi/8) < 1$. Inputs:

(I1) $D \neq 0$, $c_k \neq 0$, and $\beta_k = |\arg(c_k/D)| \leq \beta_0$ (A.13, Lemma 4.1).

(I2) $1 - g_9 \leq h \leq 1$ (Lemma 8(i) and (17)), and $1 - f_{10} \leq \cos\beta_k \leq 1$ with $f_{10} = g_9/3 + f_8^{(k)}$ (A.18, in both cases of the definition of g_9).

(I3) Lemma 6 at $t = k$: $k|c_k| = (3/2 + \rho_k)C$ with $|\rho_k| \leq f_5''$.

(I4) (14) at $t = k$ (valid since $k \leq n-1$): $\sum_{j=0}^{k-1} c_j S_{k-j} = 2D - kc_k$.

(I5) (3) at $t = k$ (valid since $k \leq n-1$): $\sum_{j=0}^{k-1} b_j S_{k-j} = c_{k-1} - kb_k$.

(I6) Corollary 2 (A.6); ($\star$) at $t = k$, in the strict form $|c_{k-1}| > B/\sqrt{2}$; Lemma 2(iii) and $\sin\alpha_k > 1/\sqrt{2}$ (A.3); (6).

By (1), $|S_{k-j}| \leq q$ for $0 \leq j \leq k-1$. The new free parameter is $\gamma > 0$, which fixes $c = 1/2 - \gamma$; the parameter ε of Corollary 2 is used once, in A.23.

A.20. **(38): f_{11}.** Let $f_{11} := \frac{3}{2} g_9 + \frac{3}{2}\sqrt{2 f_{10}} + f_5''$. Then

$$|S_k + c_1 S_{k-1} + \cdots + c_{k-1} S_1 - D/2| \leq f_{11} C. \tag{38′}$$

Proof. By (I4) the expression inside the absolute value is $2D - kc_k - D/2 = \frac{3}{2}D - kc_k$. Put $u := D/|D|$, $\psi := \arg(c_k/D)$ (so $|\psi| = \beta_k$ by (I1)) and $r := 3/2 + \rho_k$. Then $D = hCu$ and $c_k = |c_k| e^{i\psi} u$, so by (I3) $kc_k = rCue^{i\psi}$. Hence

$$\tfrac{3}{2}D - kc_k = Cu(\tfrac{3}{2}h - re^{i\psi}), \qquad \tfrac{3}{2}h - re^{i\psi} = \tfrac{3}{2}(h-1) + \tfrac{3}{2}(1 - e^{i\psi}) - \rho_k e^{i\psi}.$$

Now $|h-1| = 1 - h \leq g_9$ by (I2); $|1 - e^{i\psi}| = \sqrt{2(1-\cos\psi)} \leq \sqrt{2 f_{10}}$ by (I2), since $\cos\psi = \cos\beta_k$; and $|\rho_k e^{i\psi}| \leq f_5''$ by (I3). The triangle inequality and $|u| = 1$ give (38′). □

Biró writes (38) as $D/2 + O(f_{11}C)$. The square root comes from Lemma 8(ii): it controls $1 - \cos\beta_k$, so the angle β_k itself is controlled only to order $\sqrt{f_{10}}$. This is correct, not a gap, but costly: $f_{11} \approx \frac{3}{2}\sqrt{2g_9/3} \approx 1.22\sqrt{g_9}$ for small g_9 (float).

A.21. **(39)–(41): f_{12}, f_{13}.** Put $t_j := c_j/D$ for $0 \le j \le k-1$ (Biró's (39); $D \neq 0$ by (I1)), and

$$f_{12} := \frac{g_9}{1-g_9}, \qquad f_{13} := \frac{\sqrt{2g_9}}{1-g_9}.$$

(40′) $\sum_{j=0}^{k-1} t_j = 1$ and $0 \le \sum_{j=0}^{k-1}|t_j| - 1 \le f_{12}$. Moreover $\sum_j(|t_j| - \operatorname{Re} t_j) = \sum_j |t_j| - 1$, and every term $|t_j| - \operatorname{Re} t_j$ is ≥ 0.

Proof. $\sum_{j=0}^{k-1} c_j = D$ (as $c_0 = 1$), so $\sum t_j = 1$. Next $\sum|t_j| = C/|D| = 1/h$, and $1/h - 1 = (1-h)/h$; this is ≥ 0 since $h \le 1$, and $\le g_9/(1-g_9)$ since $1 - h \le g_9$ and $h \ge 1 - g_9 > 0$. Finally $\sum(|t_j| - \operatorname{Re} t_j) = \sum|t_j| - \operatorname{Re}\sum t_j = \sum|t_j| - 1$, and $|w| \ge \operatorname{Re} w$ for every w.

(41′) $\sum_{j=0}^{k-1}||t_j| - t_j| \le f_{13}$.

Proof. Lemma 1(v) of [Bir00b] gives $\big||t_j| - t_j\big|^2 = 2(|t_j| - \operatorname{Re} t_j)|t_j|$. By the Cauchy–Schwarz inequality and (40′),

$$\sum_j||t_j| - t_j| \le \sqrt{2}\Big(\sum_j(|t_j| - \operatorname{Re} t_j)\Big)^{1/2}\Big(\sum_j|t_j|\Big)^{1/2} = \sqrt{2(1/h - 1)/h} = \frac{\sqrt{2(1-h)}}{h} \le \frac{\sqrt{2g_9}}{1-g_9}.$$

A.22. **(42) and the set H_c: f_{14}, f_{15}, and the choice of c.** Let $f_{14} := f_{11}/(1-g_9) + qf_{13}$ and $f_{15} := \delta + qf_{12} + f_{14}$.

(42′) $\big|\sum_{j=0}^{k-1}|t_j|S_{k-j} - 1/2\big| \le f_{14}$. In particular $\sum_j|t_j|\operatorname{Re} S_{k-j} \ge 1/2 - f_{14}$.

Proof. Divide (38′) by D: $\sum t_j S_{k-j} - 1/2$ has modulus at most $f_{11}C/|D| = f_{11}/h \le f_{11}/(1-g_9)$. By (1) and (41′), $\big|\sum|t_j|S_{k-j} - \sum t_j S_{k-j}\big| \le \sum||t_j| - t_j||S_{k-j}| \le qf_{13}$. Use the triangle inequality, and $\operatorname{Re} z \ge 1/2 - |z - 1/2|$.

The set H_c. Let $\gamma > 0$ and $c := 1/2 - \gamma$ (S7). Put $H_c := \{0 \le j \le k-1 : \operatorname{Re} S_{k-j} \le c\}$ and $T_c := \sum_{j\in H_c}|t_j|$.

Claim. $(q-c)T_c \le f_{15}$, hence $T_c \le f_{15}/\gamma$.

Proof. For $j \in H_c$, $\operatorname{Re} S_{k-j} \le c$; for $j \notin H_c$, $\operatorname{Re} S_{k-j} \le |S_{k-j}| \le q$ by (1). Since $|t_j| \ge 0$,

$$\sum_j|t_j|\operatorname{Re} S_{k-j} \le q\Big(\sum_j|t_j| - T_c\Big) + cT_c = q\sum_j|t_j| - (q-c)T_c.$$

With (42′) and $\sum|t_j| \le 1 + f_{12}$ (40′), $(q-c)T_c \le q(1+f_{12}) - 1/2 + f_{14} = f_{15}$. Finally $q - c = \delta + \gamma > \gamma > 0$.

Consequence. With $y := f_{15}/\gamma$, $\sum_{j\in H_c}|c_j| \le yC$, because $|c_j| = |t_j||D| \le |t_j|C$ (as $|D| = hC \le C$).

Biró proves $(1/2 - c)T_c \le f_{15}$ (the Claim with $q - c$ replaced by the smaller $1/2 - c = \gamma$) and takes $\gamma = f_{16} := \sqrt{f_{15}}$. We keep γ free.

A.23. **(43): f_{17}.** Let $f_{17} := m_4(y) = \min\{1, (\varepsilon+\kappa)^{K_1^{(\mathrm{i})}} + \sqrt{2}(1+q)y/\varepsilon\}$, with $y = f_{15}/\gamma$. Then

$$\sum_{j\in H_c}|b_j| \le f_{17}B. \tag{43}$$

Proof. $H_c \subseteq \{0,\dots,k-1\}$, $y \ge 0$, and (7) holds with this y by A.22; apply Corollary 2. (H_c may contain $j = 0$; Corollary 2 allows this, since its part (a) counts $b_0 = 1$ inside B_s.)

A.24. **f_{18} and f_{19}.** Let $f_{18} := \sqrt{f_7 + \gamma}$ and $f_{19} := f_{18} + (q + \frac{1}{2})f_{17}$.

Claim A. For $0 \le j \le k-1$ with $j \notin H_c$: $|S_{k-j} - 1/2| < f_{18}$. For every $0 \le j \le k-1$: $|S_{k-j} - 1/2| \le q + 1/2$.

Proof. $j \notin H_c$ means $\operatorname{Re} S_{k-j} > c = 1/2 - \gamma$. By the identity $|S - 1/2|^2 = |S|^2 - \operatorname{Re} S + 1/4$ and (1), $|S_{k-j} - 1/2|^2 < q^2 - (1/2 - \gamma) + 1/4 = f_7 + \gamma$. The second statement is the triangle inequality with (1).

Claim B. $|S_k + b_1 S_{k-1} + \cdots + b_{k-1} S_1 - c_{k-1}/2| \le f_{19} B$.

Proof. Since $c_{k-1} = \sum_{j=0}^{k-1} b_j$, the expression inside the absolute value equals $\sum_{j=0}^{k-1} b_j (S_{k-j} - 1/2)$. Split the sum into $j \notin H_c$ and $j \in H_c$ and use Claim A and (43):

$$\Big|\sum_{j=0}^{k-1} b_j (S_{k-j} - \tfrac{1}{2})\Big| \le f_{18} \sum_{j \notin H_c} |b_j| + (q + \tfrac{1}{2}) \sum_{j \in H_c} |b_j| \le f_{18} B + (q + \tfrac{1}{2}) f_{17} B,$$

because $\sum_{j=0}^{k-1} |b_j| = B$.

A.25. **(44)–(47): the contradiction.**

$$\Big|\frac{c_{k-1}}{2} - k b_k\Big| \le f_{19} B. \tag{44$'$}$$

Proof. By (I5), $S_k + b_1 S_{k-1} + \cdots + b_{k-1} S_1 = c_{k-1} - k b_k$. Insert this into Claim B: $(c_{k-1} - k b_k) - c_{k-1}/2 = c_{k-1}/2 - k b_k$.

Biró's route (45)–(47). Put $f_{20} := f_{19}/(1/\sqrt{2} - 1/2) = (2 + 2\sqrt{2}) f_{19}$. Lemma 2(iii) at $t = k$ and $\sin \alpha_k > 1/\sqrt{2}$ give $|c_{k-1} - k b_k| \ge |c_{k-1}|/\sqrt{2}$. From (44$'$), since $c_{k-1} - k b_k = c_{k-1}/2 + (c_{k-1}/2 - k b_k)$, we get $|c_{k-1} - k b_k| \le |c_{k-1}|/2 + f_{19} B$. So $(1/\sqrt{2} - 1/2)|c_{k-1}| \le f_{19} B$, that is,

$$|c_{k-1}| \le f_{20} B. \tag{45}$$

By ($\star$) at $t = k$,

$$|c_{k-1}| > B/\sqrt{2}. \tag{46}$$

Together, $B/\sqrt{2} < f_{20} B$, i.e.

$$1/\sqrt{2} < f_{20}. \tag{47}$$

So if $f_{20} \le 1/\sqrt{2}$, that is, if $f_{19} \le (2 - \sqrt{2})/4$ (FIN-B), we have a contradiction.

Our final step (S8): if $f_{19} \le 1/4$ (FIN), Proposition 4.3 gives a contradiction directly from (44$'$), (46) and (6). Note that $(2 - \sqrt{2})/4 < 1/4$, so (FIN-B) implies (FIN).

A.26. **End of the proof of Theorem 3.1.** Let $q, \alpha, \beta_0, \theta, \mu, \varepsilon, \gamma$ satisfy (P1)–(P4), (L8a), (L8b) and (FIN), and suppose that some system $z_1 = 1, z_2, \ldots, z_n$ satisfies (1) with this q. Every step of A.1–A.25 applies to this system, and A.25 (with Proposition 4.3) gives a contradiction. With (FIN-B) in place of (FIN), Biró's (45)–(47) give the contradiction. None of the conditions depends on the system. Hence $\max_{1 \le j \le n} |S_j| > q$ for every system $z_1 = 1, z_2, \ldots, z_n$. □

(Strictly, (L8a) and (L8b) are not needed for the logic: without them $g_9 = 1 - \cos \beta_0$ and the chain still holds. But then f_{19} is far above $1/4$, so the theorem simply requires them.)

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